

Fauzan’s determinant construction for $\zeta(5)$

Expanded proof draft and Lean formalization specification

Original manuscript: Aabir Fauzan
Working edition prepared for Christopher D. Long

23 September 2026
Source manuscript dated 17 September 2026

Abstract

This document reconstructs and expands the argument in Aabir Fauzan’s $\zeta(5)$ *is irrational*, using the supplied 32-page PDF as the mathematical source. It preserves the determinant, rational normalization, comparison measure, and numerical constants. The local functional is given an explicit domain, bounded division and compatibility are proved, the residue-class distribution identity is expanded, and the inner determinant weights are checked by a complete source-row case analysis. All finite data in the potential and prime-sum certificates are supplied, together with an independently executed exact-arithmetic verifier. Stable declarations, external-theorem interfaces, and an autonomous formalization ledger are included. This is a mathematical working edition, not a completed Lean formalization or an author-approved revision. The original mathematical claims remain attributed to Fauzan; numerical experiments reported in the supplied audits are not used as proofs.

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1 Purpose, conventions, and the main targets

1.1 Source and editorial scope

The source is the uploaded file `ZETA5_IS_IRRATIONAL.pdf`, by Aabir Fauzan, dated 17 September 2026, 32 pages. Its SHA-256 digest is

a42c05ccc46b9a7c36428913453c4617202c59109713ba20f464ab9ec640d0da.

The original source is cited as [1]. No original publisher L^AT_EX source is assumed. Source theorem numbers are stated in the titles or surrounding text; the numbering in this working edition is independent. Authorial attribution is not transferred by this reconstruction.

Two distinctions are essential. First, the $+$ after the large parenthesis in source (5.9) is a *subscript denoting the positive part*. The indicator which follows *multiplies* that expression. The PDF does not contain an added indicator. The earlier suggestion to delete an additive indicator was a reading error, not an error in the source. We use an unambiguous piecewise definition in (5.9).

Second, $K \geq 200M^2$ is an *admissibility* condition for the explicit normalization, not an effective threshold for its asymptotic decay. For $M = 200$ it says $n \geq 200000$ when $K = 40n$; the decay assertions may require a larger, unspecified threshold. For $M = 100000$ the admissibility threshold is $n \geq 50000000000$.

1.2 Basic conventions and external inputs

Let

$$\xi = \sum_{m=1}^{\infty} \frac{1}{m^5} = \zeta(5), \quad \alpha = \frac{3}{40}, \quad \lambda = \frac{37}{40}, \quad H = \lambda + 3\alpha = \frac{23}{20}.$$

The series definition is a convenient real-valued formalization target. The identification with the real value of the analytically continued zeta function must be proved if the latter is used for the public endpoint. Bernoulli numbers have convention $B_1 = -1/2$; the standard identities used below are collected in [2]. An empty product is 1, an empty sum is 0, and $u_+ = \max(u, 0)$ for real u .

All additive valuations are normalized by $v_p(p) = 1$ and $v_p(0) = +\infty$. For $A \in \mathbb{Q}_p[X]$, $v_p^G(A)$ is the minimum coefficient valuation, with the same convention at zero. Half-integer weight inequalities can be represented in Lean by multiplying every inequality by two, rather than by changing the value group.

The following standard inputs are used with their exact conventional hypotheses: Bernoulli reflection, difference, and multiplication identities; von Staudt–Clausen; Euler’s even-zeta formula; Hermite’s Hurwitz-zeta integral; the prime number theorem in the form

$$\vartheta(T) := \sum_{p \leq T} \log p, \quad \vartheta(T)/T \longrightarrow 1; \tag{1.1}$$

and standard finite-dimensional linear algebra, complete valued-field analysis, and measure/integration results. Their needed interfaces are listed in Appendix D. No assertion is made here that a particular current Mathlib declaration already supplies any interface. The Gaussian energy identity and determinant integral used below are proved here rather than treated as unnamed external results.

Theorem 1.1 (Main polynomial target; source Theorem 2.1). *For integers $M \geq 40$ and $K \in 40\mathbb{Z}_{>0}$ satisfying $K \geq 200M^2$, the explicit polynomial $Q_{K,M}$ defined in Sections 2 and 5 belongs to $\mathbb{Z}[X]$, has degree $h = 37K/40$, and satisfies $Q_{K,M}(\xi) > 0$. There exist integers $n_{200} \geq 200000$ and $n_{100000} \geq 50000000000$ such that*

$$0 < Q_{40n,200}(\xi) < \exp(-139n^2/5) \quad (n \geq n_{200}), \quad (1.2)$$

$$0 < Q_{40n,100000}(\xi) < \exp(-7907n^2/100) \quad (n \geq n_{100000}). \quad (1.3)$$

Stable target: Zeta5.polynomial_estimates.

Lemma 1.2 (Integer-polynomial irrationality criterion). *Let $x \in \mathbb{R}$, $d, c > 0$, and let $P_n \in \mathbb{Z}[X]$ for all sufficiently large positive integers n . Suppose $\deg P_n \leq dn$ and $0 < P_n(x) < \exp(-cn^2)$. Then x is irrational.*

Stable target: Zeta5.irrational_of_integer_polynomials.

Proof. If $x = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, let D_n be an integer degree bound with $D_n \leq dn$. The evaluation identity

$$b^{D_n} P_n(a/b) = \sum_{j=0}^{D_n} [X^j] P_n a^j b^{D_n-j}$$

shows this is an integer. It is positive and smaller than $\exp(dn \log b - cn^2)$, which tends to zero. Eventually it is a positive integer less than one, a contradiction. Equivalently use $D_n = \deg P_n$, since positivity excludes the zero polynomial. $\square$

Theorem 1.3 (Fauzan's irrationality target; source Theorem 1.1). *The real number $\zeta(5)$ is irrational.*

Stable target: Zeta5.zeta_five_irrational.

Proof. Apply Theorem 1.1 and Lemma 1.2 with $d = 37$ and $c = 139/5$. The proof of the polynomial target is completed in Section 7. $\square$

Corollary 1.4 (Optional approximation target; source Corollary 1.2). *There is $b_0 \in \mathbb{Z}_{>0}$ such that $|\zeta(5) - a/b| > b^{-260}$ for all $a \in \mathbb{Z}$ and all integers $b \geq b_0$.*

Stable target: Zeta5.rational_approximation_260.

This target, proved in Appendix C, is independent of the formalization of the main irrationality endpoint: it must not delay that endpoint or be silently included among its assumptions.

2 The functional, determinant, degree, and positivity

Throughout, choose $n \in \mathbb{Z}_{>0}$ and put

$$K = 40n, \quad N = 3n, \quad h = 37n = K - N. \quad (2.1)$$

Define

$$D_m(t) = \prod_{j=1}^m (t + j^2), \quad D_0 = 1, \quad H_j^{(5)} = \sum_{v=1}^j v^{-5}, \quad H_0^{(5)} = 0.$$

Let $\mathcal{R}_-$ be the $\mathbb{Q}$ -vector space of rational functions in t whose poles, if any, are simple and lie among $-j^2$, $j \geq 1$. A function in this space has a unique polynomial part and a unique finite simple partial-fraction expansion.

Definition 2.1 (The rational functional). The $\mathbb{Q}$ -linear map $\mu_X : \mathcal{R}_- \rightarrow \mathbb{Q}[X]$ is defined by

$$\mu_X(t^e) = \frac{(-1)^e B_{2e+2}(2e+3)(2e+4)(2e+5)}{24} \quad (e \geq 0), \quad (2.2)$$

$$\mu_X\left(\frac{1}{t+j^2}\right) = j^4(X - H_j^{(5)}) - \frac{1}{4} + \frac{1}{2j} \quad (j \geq 1). \quad (2.3)$$

Its restriction to polynomials is denoted μ .

Stable target: `Zeta5.muX`.

The output is affine in X . The domain is a vector space, not a ring closed under unrestricted multiplication: repeated poles are not silently allowed.

Set

$$G_K(X) = \left[\mu_X \left(\frac{D_N(t)^6 t^{i+j}}{D_K(t)} \right) \right]_{0 \leq i, j < h}, \quad \Delta_K(X) = \det G_K(X), \quad (2.4)$$

$$S_K = \frac{(K!)^{2h} 4^{h-1}}{(N!)^{12h} \prod_{i=1}^{h-1} ((2i)!)^2}, \quad F_K = S_K \Delta_K. \quad (2.5)$$

After cancelling D_N , the rational input is $W(t)t^{i+j}/D_{\text{tail}}(t)$, where $W = D_N^5$ and $D_{\text{tail}} = D_K/D_N$. Its h possible poles are distinct, so Definition 2.1 applies. Section 5 defines a positive rational $m_{K,M}$ and

$$Q_{K,M} = m_{K,M} F_K. \quad (2.6)$$

Proposition 2.2 (Exact degree; source (2.9)). *The polynomial Δ_K has degree h , and*

$$[X^h] \Delta_K = (-1)^{h(h-1)/2} \prod_{j=N+1}^K j^4 D_N(-j^2)^5 \neq 0. \quad (2.7)$$

Every nonzero scalar multiple, including F_K and $Q_{K,M}$ when defined, has the same degree.

Stable target: `Zeta5.determinant_degree`.

Proof. Write $s_j = -j^2$, $N < j \leq K$, and $V_{ij} = s_j^i$. Extracting the coefficient of X from each entry gives

$$[X]G_K = V \operatorname{diag} \left(\frac{j^4 D_N(s_j)^5}{D'_{\text{tail}}(s_j)} \right) V^T.$$

The coefficient of X^h in a determinant of affine entries is the determinant of this coefficient matrix. Since

$$\det(V)^2 = \prod_{j < k} (s_k - s_j)^2, \quad \prod_j D'_{\text{tail}}(s_j) = (-1)^{h(h-1)/2} \prod_{j < k} (s_k - s_j)^2,$$

the Vandermonde factors cancel and give (2.7). All remaining factors are nonzero. The degree is at most h by multilinearity, and hence is exactly h . $\square$

Proposition 2.3 (Positive moment representation; source Proposition 2.2). *Put*

$$w(y) = \frac{(2\pi)^4 y^5}{12} \sum_{\ell=1}^{\infty} \ell^4 e^{-2\pi\ell y} \quad (y > 0). \quad (2.8)$$

For every $R \in \mathcal{R}_-$,

$$\mu_{\xi}(R) = \int_0^{\infty} R(y^2) w(y) dy. \quad (2.9)$$

The integral is absolutely convergent. The matrix $G_K(\xi)$ is real symmetric positive definite; in particular $\Delta_K(\xi) > 0$.

Stable target: `Zeta5.positive_moment_representation`.

Proof. The weight is positive. The identity $\sum_{\ell \geq 1} \ell^4 q^{\ell} = q(1 + 11q + 11q^2 + q^3)/(1 - q)^5$ shows that $w(y) \rightarrow 1/\pi$ as $y \downarrow 0$ and $w(y) = O((1 + y)^5 e^{-2\pi y})$ as $y \rightarrow \infty$. Thus each polynomial or permitted pole term is absolutely integrable; the same holds for their finite sums.

For $e \geq 0$, monotone convergence and the gamma integral give

$$\int_0^{\infty} y^{2e} w(y) dy = \frac{(2e + 5)!}{12(2\pi)^{2e+2}} \zeta(2e + 2),$$

which is (2.2) by Euler's formula.

For $a > 0$, let $f(y) = (e^{2\pi y} - 1)^{-1}$. Termwise differentiation on compact subintervals of $(0, \infty)$ gives $w(y) = y^5 f^{(4)}(y)/12$. An elementary differentiation identity is

$$\frac{d^4}{dy^4} \frac{y^5}{y^2 + a^2} = 24a^4 \operatorname{Im}(a - iy)^{-5}.$$

Integrate by parts four times first on $[\varepsilon, R]$. At zero the boundary term involving the k th derivative of $y^5/(y^2 + a^2)$ and the $(3 - k)$ th derivative of f , for $0 \leq k \leq 3$, is $O(y^{5-k} y^{-4+k}) = O(y)$. At infinity it tends to zero exponentially. The differentiated integrands are absolutely integrable, so passage to the endpoints is justified. Hermite's integral formula at $s = 5$ now gives

$$\int_0^{\infty} \frac{w(y)}{y^2 + a^2} dy = 2a^4 \int_0^{\infty} \frac{\operatorname{Im}(a - iy)^{-5}}{e^{2\pi y} - 1} dy = a^4 \zeta(5, a) - \frac{1}{2a} - \frac{1}{4}.$$

At $a = j \in \mathbb{Z}_{>0}$, use $\zeta(5, j) = \xi - H_{j-1}^{(5)}$ and $H_j^{(5)} - H_{j-1}^{(5)} = j^{-5}$ to recover (2.3). This verifies (2.9) on the partial-fraction basis and hence on its domain.

Finally, a nonzero real polynomial q of degree less than h satisfies

$$\mathbf{q}^T G_K(\xi) \mathbf{q} = \int_0^{\infty} \frac{D_N(y^2)^6 q(y^2)^2}{D_K(y^2)} w(y) dy > 0.$$

The integrand is continuous and positive on some interval: q cannot vanish on all positive real numbers. This proves positive definiteness and the determinant assertion. $\square$

3 The local rational functional: explicit domains and distribution

3.1 Polynomial and integer-pole definitions

Let $L(x^k) = B_k$ on $\mathbb{Q}[x]$, and let $\tau(P) = L(P''')/24$. For $r \in \mathbb{Z}$, define the nonnegative integer

$$d(r) = \begin{cases} r, & r \geq 0, \\ -r-1, & r < 0, \end{cases} \quad \tau_X\left(\frac{1}{x-r}\right) = H_{d(r)}^{(5)} - X.$$

Extend by polynomial division and finite simple partial fractions to the vector space of rational functions with polynomial part and simple integer poles. At a fixed prime, extend all these rational functionals coefficientwise from $\mathbb{Q}$ to $\mathbb{Q}_p$ before applying them to p -adic inputs. The following identities include the required domain restriction:

$$\mu_X(R) = \tau_X(x^5 R(-x^2)), \quad (3.1)$$

$$\tau_X(g(-1-x)) = -\tau_X(g(x)), \quad (3.2)$$

$$\tau_X(g(x+1) - g(x)) = g^{(4)}(0)/24 \quad \text{if } g \text{ is regular at } 0. \quad (3.3)$$

For (3.1), monomials follow by differentiating three times. For a pole use

$$\frac{x^5}{j^2 - x^2} = -x^3 - j^2 x - \frac{j^4}{2} \left(\frac{1}{x-j} + \frac{1}{x+j} \right),$$

then $d(j) = j$ and $d(-j) = j-1$. The polynomial cases of (3.2)–(3.3) follow from Bernoulli reflection and difference. For the pole cases use $d(-1-r) = d(r)$ and, for $r \neq 0$,

$$H_{d(r-1)}^{(5)} - H_{d(r)}^{(5)} = -r^{-5}.$$

The last identity holds for negative as well as positive r .

3.2 Restricted power series and meromorphic direct sums

Fix a prime $p \geq 7$. Let

$$\mathcal{A} = \mathbb{Q}_p\langle z \rangle = \{ \sum_{d \geq 0} f_d z^d : f_d \in \mathbb{Q}_p, f_d \rightarrow 0 \}, \quad \|f\| = \sup_d |f_d|_p.$$

Completeness follows by coefficientwise completeness and uniform control of the tails. Polynomial truncations are dense. Put

$$\kappa_d = \frac{d(d-1)(d-2)B_{d-3}}{24} \quad (d \geq 3), \quad \kappa_0 = \kappa_1 = \kappa_2 = 0.$$

Von Staudt–Clausen implies $v_p(\kappa_d) \geq -1$ for every d , and $\kappa_d \in \mathbb{Z}_p$ for $d \leq p+1$. Hence

$$\tau\left(\sum_{d \geq 0} f_d z^d\right) = \sum_{d \geq 0} f_d \kappa_d, \quad \|\tau\| \leq p. \quad (3.4)$$

The series converges because its terms tend to zero in the complete nonarchimedean field.

For a finite set S of distinct integers, let

$$\mathcal{B}_S = \mathcal{A} \oplus \bigoplus_{r \in S} \mathbb{Q}_p \frac{1}{z-r}.$$

Give this space the maximum of the analytic coefficient norm and residue norms. It embeds into the localization of $\mathcal{A}$ obtained by inverting $\prod_{r \in S} (z - r)$. Indeed, if $f + \sum_r c_r / (z - r) = 0$, multiplication by that product and evaluation at r gives $c_r \prod_{s \neq r} (r - s) = 0$, so each residue vanishes and then $f = 0$. This proves uniqueness of the decomposition. No claim that $\mathcal{B}_S$ itself is a ring is needed.

For $Y \in \mathbb{Q}_p[X]$ define

$$\tau_Y^{\text{ext}} \left(f + \sum_{r \in S} \frac{c_r}{z - r} \right) = \tau(f) + \sum_{r \in S} c_r (H_{d(r)}^{(5)} - Y). \quad (3.5)$$

These maps are compatible when S is enlarged. Their images lie in the finite-dimensional space $\mathbb{Q}_p + \mathbb{Q}_p Y \subset \mathbb{Q}_p[X]$, which is complete for the Gauss norm. In particular, convergence in this proof never relies on completeness of the entire unbounded-degree polynomial ring.

Lemma 3.1 (Bounded division in the restricted power-series space). *Let $T(z) = \prod_{\nu=1}^t (z - r_\nu)$, where $r_\nu \in \mathbb{Z}$ and every $r_\nu - r_\eta$, $\nu \neq \eta$, is a p -adic unit. For $f \in \mathcal{A}$,*

$$\frac{f}{T} = P + \sum_{\nu=1}^t \frac{c_\nu}{z - r_\nu}, \quad c_\nu = \frac{f(r_\nu)}{T'(r_\nu)},$$

with $P \in \mathcal{A}$ and $\max(\|P\|, |c_1|_p, \dots, |c_t|_p) \leq \|f\|$. For a polynomial f this is ordinary division and partial fractions; its polynomial quotient has degree at most $\deg f$.

Stable target: Zeta5.Local.bounded_division.

Proof. Evaluation at $r \in \mathbb{Z}_p$ has norm at most one. More explicitly, if $f = \sum f_k z^k$, then

$$\frac{f(z) - f(r)}{z - r} = \sum_{n \geq 0} \left(\sum_{k \geq n+1} f_k r^{k-1-n} \right) z^n.$$

The inner series converges, its absolute value is at most $\sup_{k \geq n+1} |f_k|_p$, and these bounds tend to zero. This proves bounded division by one linear factor after removing the remainder.

For polynomial f , division by the monic integral polynomial T and repeated subtraction of its leading term give a quotient and remainder with coefficients bounded by $\|f\|$. Since $T'(r_\nu)$ is a unit, the partial-fraction residues have the same bound. Apply this polynomial map to the truncations of an arbitrary f and use completeness of $\mathcal{B}_S$. The displayed evaluation formula for c_ν passes to the limit. The localization identity passes to the limit after multiplication by T ; uniqueness was proved above. $\square$

Proposition 3.2 (Integral extension with a degree-restricted leading term; source Lemma 3.1). *Under the hypotheses of Lemma 3.1, suppose also $d(r_\nu) < p$, $Y \in \mathbb{Z}_p[X]$, and $U_j \in \mathbb{Z}_p[z]$ for every $j \geq 0$. If $\deg U_0 \leq p + 1$, then the series below converges in $\mathcal{B}_S$ and*

$$\tau_Y^{\text{ext}} \left(\sum_{j \geq 0} \frac{p^j U_j(z)}{T(z)} \right) \in \mathbb{Z}_p[X]. \quad (3.6)$$

Stable target: Zeta5.Local.integral_extension.

Proof. Lemma 3.1 bounds the norm of each U_j/T by one; the factors p^j force convergence. Each principal-part value $H_{d(r_\nu)}^{(5)} - Y$ is integral because $d(r_\nu) < p$. The analytic functional loses at most one factor of p , by (3.4). For $j = 0$, the quotient is a polynomial of degree at most $p + 1$, so its value is integral without that loss. For $j \geq 1$, p^j absorbs it. All partial sums have integral output. Their limit in $\mathbb{Q}_p + \mathbb{Q}_p Y$ is integral coefficientwise because $\mathbb{Z}_p$ is closed. The case $T = 1$ is included. $\square$

Lemma 3.3 (Far poles and order of expansion). *If $s \in \mathbb{Q}_p$ has $v_p(s) < 0$, then $(z - s)^{-1} \in \mathcal{A}$ and*

$$\tau^{\text{an}}\left(\frac{1}{z - s}\right) = -\frac{1}{4} \sum_{k \geq 0} \binom{k+3}{3} B_k s^{-k-4}. \quad (3.7)$$

If a rational function has polynomial part and only simple poles, each pole being either an integer or outside $\mathbb{Z}_p$, its value under (3.5) is independent of whether one first takes global partial fractions or expands its far denominator factors in $\mathcal{A}$ and then separates its near principal parts.

Stable target: Zeta5.Local.far_pole_compatibility.

Proof. The geometric series $(z - s)^{-1} = -\sum_{d \geq 0} s^{-d-1} z^d$ converges in the coefficient norm. Apply the continuous functional and substitute $d = k + 3$ to obtain (3.7). Products of the far expansions converge in $\mathcal{A}$ and represent the corresponding rational functions there. Both proposed procedures therefore give the same element of the localization of $\mathcal{A}$ at the near-pole factors. Both have the same simple principal parts and analytic remainder by uniqueness. Applying the single map (3.5) proves the assertion. No use of the integer-pole harmonic formula at a nonintegral far pole is permitted. $\square$

Lemma 3.4 (Reflection and difference on the completed domain). *On the union of the spaces $\mathcal{B}_S$, for finite integer pole sets S , reflection satisfies*

$$\tau_Y^{\text{ext}}(g(-1 - z)) = -\tau_Y^{\text{ext}}(g(z)).$$

If g is regular at zero, translation satisfies

$$\tau_Y^{\text{ext}}(g(z + 1) - g(z)) = g^{(4)}(0)/24.$$

Both assertions include analytic rational functions with far poles via Lemma 3.3.

Stable target: Zeta5.Local.completed_translation_reflection.

Proof. The substitutions $z \mapsto z + 1$ and $z \mapsto -1 - z$ are bounded maps of $\mathcal{A}$ into itself: the binomial coefficients are integral and the polynomial substitution maps have norm at most one, so they extend from dense polynomial truncations. Differentiation and evaluation at zero are also continuous. Thus the polynomial identities (3.2)–(3.3) extend to the analytic part. For an integer principal part reflection sends its pole to $-1 - r$, and translation sends it to $r - 1$. The harmonic identities already checked prove the result on these terms. In the difference identity the term $r = 0$ is excluded exactly by regularity at zero. Linearity combines the parts. A far pole remains far under these unit translations and reflection, so its analytic interpretation is consistent throughout. $\square$

3.3 Distribution, with the pole recurrence made explicit

Define

$$C_p = \sum_{a=1}^{p-1} \tau^{\text{an}}\left(\frac{1}{z + a/p}\right), \quad Y = p^5 X + C_p. \quad (3.8)$$

Lemma 3.5 (Integrality of the distribution parameter). *One has $C_p \in p^5\mathbb{Z}_p$, and hence $Y \in \mathbb{Z}_p[X]$. Stable target: `Zeta5.Local.distribution_parameter_integral`.*

Proof. The $k = 0$ terms of (3.7) sum to $-p^4 \sum_{a=1}^{p-1} a^{-4}/4$. Modulo p this sum of fourth inverse powers is zero: $p-1 \nmid 4$ for $p \geq 7$, and the multiplicative group of $\mathbb{F}_p$ gives the standard power-sum identity. The $k = 1$ terms have valuation at least 5. Each $k \geq 2$ term has valuation at least $k+3 \geq 5$, using $v_p(B_k) \geq -1$. The series converges, so the claimed closed ideal contains its sum. $\square$

Proposition 3.6 (Full distribution identity; source Lemma 3.2). *Let $g \in \mathbb{Q}_p(z)$ have polynomial part and finitely many simple integer poles. With Y as in (3.8),*

$$\tau_X(g) = p^{-4} \sum_{a=0}^{p-1} \tau_Y^{\text{ext}}(g(a + pz)) \quad \text{in } \mathbb{Q}_p[X]. \quad (3.9)$$

In each summand a pole $(r-a)/p$ is an integer when $r \equiv a \pmod{p}$, and has valuation -1 otherwise; it is interpreted accordingly. No restriction $d(r) < p$ is imposed on the original integer poles in this proposition. That restriction belongs only to the integrality estimate.

Stable target: `Zeta5.Local.distribution`.

Proof. All summands are in the completed domain by Lemma 3.3. Denote the right side by $T(g)$. For a polynomial, Bernoulli multiplication gives $\sum_{a=0}^{p-1} L(P(a + pz)) = pL(P)$. Three differentiations contribute a further factor p^3 , hence $T(P) = \tau(P)$. These identities are rational polynomial identities and therefore hold for coefficients in $\mathbb{Q}_p$ as well.

For $g_0(z) = 1/z$, definition (3.8) gives

$$T(g_0) = p^{-5}(-Y + C_p) = -X.$$

Next, replacing a by $p-1-a$ and z by $-1-z$ in the sum, and applying Lemma 3.4, gives $T(g(-1-z)) = -T(g(z))$. If g is regular at zero, telescoping the finite sum gives

$$\begin{aligned} T(g(z+1) - g(z)) &= p^{-4} \tau_Y^{\text{ext}}(g(p(z+1)) - g(pz)) \\ &= p^{-4} \frac{(g(pz))^{(4)}|_{z=0}}{24} = \frac{g^{(4)}(0)}{24}. \end{aligned}$$

Here $g(pz)$ belongs to the completed domain and remains regular at zero; thus the completed difference identity applies with precisely its stated hypothesis.

For $r > 0$ let $g_r(z) = 1/(z-r)$. Since g_r is regular at zero,

$$T(g_{r-1}) - T(g_r) = -r^{-5}.$$

Induction from $T(g_0) = -X$ yields $T(g_r) = H_r^{(5)} - X$ for every $r \geq 0$. Reflection gives $T(g_{-1-r}) = T(g_r)$ and hence all negative integer poles. Agreement on polynomials and simple principal parts proves (3.9). This argument is independent of any truncated p -adic numerical test. $\square$

3.4 Integer-valued polynomials and the small-prime estimate

For a prime p and integer $D \geq 1$, write

$$\ell_p(D) = \max\{e \in \mathbb{N} : p^e \leq D\} = \lfloor \log_p D \rfloor.$$

The integer characterization is preferable in a formal development.

Lemma 3.7 (Losses for integer-valued polynomials). *Let $A \in \mathbb{Q}_p[x]$ have degree at most d and map $\mathbb{Z}_p$ into $\mathbb{Z}_p$. Then its coefficients in the basis $\binom{x}{k}$ are integral and*

$$v_p(A(x) - A(y)) \geq v_p(x - y) - \ell_p(\max(1, d)) \quad (x, y \in \mathbb{Z}_p).$$

Its r th derivative is integer-valued up to a loss of at most $r\ell_p(\max(1, d))$. If $P(\mathbb{Z}_p) \subseteq p^{-A_0}\mathbb{Z}_p$ and $\deg P \leq d$, then

$$v_p(\tau(P)) \geq -A_0 - 4\ell_p(d + 1) - v_p(24). \quad (3.10)$$

Stable target: Zeta5.Local.integer_valued_estimates.

Proof. The binomial coefficients of A are its finite forward differences at zero, hence integral. Vandermonde's identity, together with $\binom{z}{j} = z\binom{z-1}{j-1}/j$, gives the difference estimate. For derivatives expand $\binom{x+z}{k}$ by Vandermonde. The coefficient of z^r in $\binom{z}{j}$ is a sum of products of r reciprocals of positive integers at most j : use $\binom{z}{j} = (-1)^{j-1}(z/j) \prod_{u=1}^{j-1}(1 - z/u)$. Each product has valuation at least $-r\ell_p(\max(1, d))$; multiplication by $r!$ cannot worsen it.

The formal generating identity $L((1+z)^x) = \log(1+z)/z$ gives $L(\binom{x}{k}) = (-1)^k/(k+1)$. Apply the third-derivative estimate, expand the resulting polynomial in the binomial basis, and apply this last identity. A further loss at most $\ell_p(d+1)$ and the factor 24 give (3.10). Scaling by p^{A_0} treats the stated range. $\square$

Proposition 3.8 (Small-prime rational estimate; source Lemma 3.3). *For any prime p , let $A \in \mathbb{Q}_p[x]$ be integer-valued on $\mathbb{Z}_p$ with $\deg A \leq d$, and let $K \geq 1$. Set*

$$g(x) = \frac{(K!)^2 A(x)}{\prod_{-K \leq r \leq K, r \neq 0} (x - r)}.$$

Then

$$v_p^G(\tau_X(g)) \geq -6\ell_p(\max(2K, d + 1)) - v_p(24). \quad (3.11)$$

Stable target: Zeta5.Local.small_prime_functional_bound.

Proof. Let $S = \{-K, \dots, -1, 1, \dots, K\}$ and $L_0 = \ell_p(2K)$. Write $g = P + \sum_{r \in S} c_r/(x - r)$. The residue is, up to sign,

$$c_r = \frac{(K!)^2 r A(r)}{(K+r)!(K-r)!}, \quad v_p(c_r) \geq -L_0.$$

For each power $p^a \leq 2K$ the difference between the two denominator factorial floor sums and $2 \lfloor K/p^a \rfloor$ is at most one. Larger powers contribute zero, which proves the bound.

Suppose first x lies outside every ball $r + p^{L_0+1}\mathbb{Z}_p$. At level p^a , $a \leq L_0$, each of the two length- K integer intervals in S contains at most $\lfloor K/p^a \rfloor + 1$ elements congruent to x . There are no roots

with $v_p(x - r) > L_0$. Counting by levels therefore gives $v_p(g(x)) \geq -2L_0$. Each principal part has the same lower bound, so $v_p(P(x)) \geq -2L_0$.

Now suppose $x \in r + p^{L_0+1}\mathbb{Z}_p$. This center is unique, since distinct roots differ by an integer of absolute value at most $2K$. Put

$$F_r(x) = \frac{(K!)^2}{\prod_{s \in S \setminus \{r\}} (x - s)}.$$

For every $s \neq r$, $v_p(x - s) = v_p(r - s) \leq L_0$. Thus $v_p(F_r(x)) = v_p(F_r(r)) \geq -L_0$. If $q = v_p(x - r) > L_0$, expand the finite product

$$\frac{F_r(x)}{F_r(r)} = \prod_{s \neq r} \left(1 + \frac{x - r}{r - s}\right)^{-1}.$$

Every factor is 1 modulo p^{q-L_0} , whence $v_p(F_r(x) - F_r(r)) \geq q - 2L_0$. Subtracting the principal part at r yields

$$\frac{A(x)F_r(x) - A(r)F_r(r)}{x - r} = \frac{(A(x) - A(r))F_r(x)}{x - r} + \frac{A(r)(F_r(x) - F_r(r))}{x - r}.$$

Lemma 3.7 bounds its valuation below by $-L_0 - \max(L_0, \ell_p(\max(1, d)))$. All other principal parts are bounded below by $-2L_0$. Continuity of the polynomial P includes the centers. Consequently

$$P(\mathbb{Z}_p) \subseteq p^{-L_0-M_0}\mathbb{Z}_p, \quad M_0 = \max(L_0, \ell_p(\max(1, d))).$$

Here M_0 is local to this proof, not the real-potential constant used later. Since $\deg P \leq d$, (3.10) gives the required $-6\ell_p(\max(2K, d+1)) - v_p(24)$ bound for its value. For a principal part, $d(r) \leq K$ implies $v_p(H_{d(r)}^{(5)}) \geq -5\ell_p(K)$, and its coefficient of X has valuation at least $-L_0$. This gives the same bound coefficientwise. The ultrametric inequality finishes the proof. $\square$

Define

$$q_0(t) = 1, \quad q_i(t) = \frac{(-1)^i 2t D_{i-1}(t)}{(2i)!} \quad (i \geq 1), \quad f_i(t) = \left(\frac{D_N(t)}{(N!)^2}\right)^3 q_i(t).$$

The identities

$$q_i(-x^2) = \binom{x+i}{2i} + \binom{x+i-1}{2i}, \quad \frac{D_N(-x^2)}{(N!)^2} = \binom{N-x}{N} \binom{N+x}{N}$$

show that $f_i(-x^2)$ is integer-valued on every $\mathbb{Z}_p$. The triangular change from monomials to q_i gives exactly

$$F_K(X) = \det \left[(K!)^2 \mu_X \left(\frac{f_i(t) f_j(t)}{D_K(t)} \right) \right]_{0 \leq i, j < h}. \quad (3.12)$$

The pullback numerator $x^5 f_i(-x^2) f_j(-x^2)$ has degree at most $12N + 4h + 1$, so $\max(2K, d+1) \leq 5K$. Applying Proposition 3.8 entrywise and expanding the determinant proves the universal bound

$$v_p^G(F_K) \geq -6h\ell_p(5K) - hv_p(24) \quad \text{for every prime } p. \quad (3.13)$$

Stable target: `Zeta5.small_prime_determinant_bound`.

4 Local determinant estimates

Fix an integer $M \geq 40$ and assume $K \in 40\mathbb{Z}_{>0}$ and $K \geq 200M^2$. The small-prime estimate is used for $p \leq K/M$. Two separate bases give stronger estimates above this cutoff.

4.1 The inner range: allocation, degree checks, and every weight case

Suppose

$$K/M < p \leq K/3. \quad (4.1)$$

Then $p > 200M$, $p^2 > 5K$, and $p \geq 7$. Put $m = (p-1)/2$, and for $1 \leq a \leq m$ define

$$\ell_A(a) = \#\{1 \leq j \leq A : j \equiv a \text{ or } -a \pmod{p}\}, \quad m_A = \lfloor A/p \rfloor.$$

If a row polynomial has a zero of order $\nu_i(a)$ at $t = -a^2$, the residue substitutions $x = c + pz$ in (3.1) and (3.9) give the following candidate entry bounds:

$$\nu_i(a) + \nu_j(a) + 6\ell_N(a) - \ell_K(a) - 4 \quad (c = a \text{ or } p-a), \quad (4.2)$$

$$2\nu_i(0) + 2\nu_j(0) + 12m_N - 2m_K + 1 \quad (c = 0). \quad (4.3)$$

We justify the degree and completed-domain conditions after specifying the basis; the displayed powers alone are not yet the full argument.

Set $L_0 = 4M + 10$ and $b_a = 3\ell_N(a)$. There are unique integers T, E with

$$mT + E = h - L_0 + 3(N - m_N), \quad 0 \leq E < m. \quad (4.4)$$

Order the ordinary classes by nonincreasing $\ell_K(a)$, breaking ties by increasing a , and give $\epsilon_a = 1$ to the first E classes and 0 to the others. Define

$$L_a = T - b_a + \epsilon_a, \quad Z_a = L_a + b_a = T + \epsilon_a.$$

Because $\sum_{a=1}^m \ell_N(a) = N - m_N$, $L_0 + \sum_{a=1}^m L_a = h$. With $x = K/p \in [3, M)$, (4.4) implies

$$2Hx - \frac{21}{20} < T < 2Hx, \quad b_a \leq 6\alpha x + 3.$$

For the lower bound, write the numerator in (4.4) as $HK - L_0 - 3m_N$, use $2E/(p-1) < 1$, and use $(2L_0 + 6m_N)/(p-1) < 1/20$. For the upper bound, $2L_0 + 6m_N > 2Hx$. These inequalities follow from $p > 200M$, $m_N \leq \alpha M$, and $L_0 = 4M + 10$. Consequently

$$T - b_a > 2\lambda x - \frac{81}{20} \geq \frac{3}{2},$$

so each ordinary L_a is a positive integer.

For $0 \leq a \leq m$ and $0 \leq i < L_a$, let

$$E_{a,i}(t) = \prod_{\substack{0 \leq c \leq m \\ c \neq a}} (t + c^2)^{L_c} (t + a^2)^i. \quad (4.5)$$

All degrees are less than h . The factors $t + c^2$ are pairwise coprime modulo p . In the Chinese remainder decomposition for their powers, the local blocks of (4.5) are triangular with unit diagonal. Thus these h polynomials form a $\mathbb{Z}_p$ -unimodular basis of polynomials of degree less than h .

Define half-integer weights

$$w_{a,i} = i + b_a - \frac{\ell_K(a) + 4}{2} \quad (a > 0), \quad (4.6)$$

$$w_{0,i} = \min \left(2i + 6m_N - m_K + \frac{1}{2}, \min_{c>0} \left(Z_c - \frac{\ell_K(c) + 4}{2} \right) \right), \quad (4.7)$$

$$\gamma_p^{\text{in}} = 2 \sum_{a=0}^m \sum_{i=0}^{L_a-1} w_{a,i} \in \mathbb{Z}. \quad (4.8)$$

Proposition 4.1 (Inner valuation; source Proposition 4.1). *Under (4.1), $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$.*

Stable target: Zeta5. inner_determinant_valuation.

Proof. The local summands really have the required integral form. After $x = c + pz$, a near denominator factor is p times a linear factor with an integer zero. The near integers have absolute value at most $M + 1$ and mutual nonzero differences of absolute value less than p , so those differences are units. Their $d(r)$ indices are less than p . A far factor is a unit times a geometric series in pz . After extracting the indicated power of p , the ordinary summand has the form $\sum_{j \geq 0} p^j U_j / T$ of Proposition 3.2. Its leading numerator has degree at most

$$\nu_i(a) + \nu_j(a) + 6\ell_N(a) \leq 2(T + 1) < \frac{23}{5}M + 2 < p + 1.$$

Here every row satisfies $\nu_i(a) + b_a \leq Z_a \leq T + 1$. At the zero source, x^5 contributes $p^5 z^5$, a zero of order ν in t contributes $p^{2\nu}$, and the $\pm jp$ poles contribute the $2m_K$ denominator factors. Its leading numerator degree is at most

$$5 + 2\nu_i(0) + 2\nu_j(0) + 12m_N \leq 5 + 4L_0 + 12m_N \leq \frac{169}{10}M + 45 < p + 1.$$

The outside distribution factor is p^{-4} . Together with $Y \in \mathbb{Z}_p[X]$, these observations prove (4.2)–(4.3) for the *whole* summands, including their analytic polynomial quotients.

Each source dominates the assigned weight for each row. The following table lists the half-bound contributed by one row. Adding the two row entries in the same source column gives the entry bound.

Row	Ordinary source $c > 0$	Zero source
$(a, i), a = c > 0$	$w_{a,i}$	$2L_0 + 6m_N - m_K + 1/2$
$(a, i), a \neq c, a > 0$	$Z_c - (\ell_K(c) + 4)/2$	$2L_0 + 6m_N - m_K + 1/2$
$(0, i)$	$Z_c - (\ell_K(c) + 4)/2$	$2i + 6m_N - m_K + 1/2$

The zero row is covered directly by the minimum in (4.7). For an ordinary row $a \neq c$, since $i \leq L_a - 1$, the difference between the table entry and its assigned weight is at least

$$Z_c - Z_a + 1 + \frac{\ell_K(a) - \ell_K(c)}{2}.$$

The counts ℓ_K take two consecutive values. The exhaustive extra-row cases are

ϵ_a	ϵ_c	Lower bound for the difference
0	0	1/2
1	1	1/2
0	1	3/2
1	0	0, because the allocation order gives $\ell_K(a) \geq \ell_K(c)$

At the zero source an ordinary row has half-bound at least $2L_0 - m_K + 1/2 \geq 7M + 41/2$. On the other hand $w_{a,i} < T + 1 \leq 2HM + 1$. The zero source therefore also dominates. These checks exhaust every row and every source.

Consequently every matrix entry in this basis has Gauss valuation at least the sum of its two assigned row weights. In a determinant term those weights occur twice in total, so its valuation is at least γ_p^{in} . The ultrametric inequality gives the determinant bound. A unimodular basis change multiplies its determinant by a square of a unit and does not alter the valuation. $\square$

4.2 The outer range: polynomial correction and its actual rank

Assume for this subsection that

$$p \geq 7, \quad p \leq K < 3p, \quad p^2 > 2K, \quad 2N < p, \quad 5N \leq 2p - 2. \quad (4.9)$$

These conditions hold for $K/3 < p \leq K$ under the global admissibility hypotheses. Formula (2.2) gives $v_p(\mu(t^e)) \geq -1$. It gives integrality for $e < 2p - 3$: the only possible positive multiples of $p - 1$ among $2e + 2$ are the first three, whose denominators are cancelled respectively by $2e + 3$, $2e + 4$, and $2e + 5$.

Choose $c_e \in \mathbb{Z}_p$ with $c_e \equiv p\mu(t^e) \pmod{p}$, choosing $c_e = 0$ for $e < 2p - 3$. Define $\mu_0(t^e) = \mu(t^e) - c_e/p$ and leave all the simple-pole values unchanged. This gives a matrix decomposition

$$G_K = A + p^{-1}L, \quad L \in M_h(\mathbb{Z}_p), \quad \text{rank}_{\mathbb{Q}_p} L \leq r_p := \max(0, K + 4N - 2p + 2). \quad (4.10)$$

Indeed, the polynomial quotient of $W(t)t^{i+j}/D_{\text{tail}}(t)$ has degree at most $i + j + 6N - K$. Therefore $L_{ij} = 0$ if $i + j < K - 6N + 2p - 3$. If $r_p = 0$, all entries vanish. Otherwise $h - r_p = 2p - 5N - 2 \geq 0$; every row with $i < h - r_p$ vanishes, as does every such column. This proves rank over the field, not merely rank modulo p . Polynomial division is by a monic integral polynomial, so the entries of L are integral before and after a unimodular basis change.

For an ordinary square class $a \in \{1, \dots, (p-1)/2\}$, let $\ell = \ell_K(a)$ and $\delta = \mathbf{1}_{a \leq N}$. There are $\ell - \delta$ remaining poles in that class. Let Q_a be the product of their factors, $P_a = D_{\text{tail}}/Q_a$, and E_a the product of the factors corresponding to $j > p$ in that class. There are exactly $\ell - 2$ such factors. Use the rows $P_a q_{a,i}$, where

$$q_{a,i}(t) = \begin{cases} (t + a^2)^i, & 0 \leq i < \ell - 2, \\ E_a(t)(t + a^2)^{i-(\ell-2)}, & \ell - 2 \leq i < \ell - \delta. \end{cases} \quad (4.11)$$

For the zero class use P_0 times 1, or $1, t + p^2$, according as the remaining poles are p , or $p, 2p$. Within each class the local polynomials are monic of successive degrees. Distinct class resultants are units, so Chinese remainders again give a unimodular basis.

Cross-class entries for A are integral: their rational inputs cancel to polynomials with integral coefficients, and all moments of μ_0 are integral. Within an ordinary class, if either row index is at

least $\ell - 2$, all poles with $j > p$ cancel. At most the poles $a, p - a$ remain. Both values in (2.3) are integral and congruent modulo p . For their constant coefficients use

$$H_{p-a}^{(5)} \equiv H_{a-1}^{(5)} \pmod{p},$$

which follows by reflecting the finite harmonic sum and using $\sum_{v=1}^{p-1} v^{-5} = 0$ in $\mathbb{F}_p$ for $p \geq 7$. The terms $1/(2a)$ and $1/(2(p-a))$, together with the harmonic increment a^{-5} , give exactly the required congruence. The coefficients of X are also congruent since $a^4 \equiv (p-a)^4$.

For precision, if two nodes u, v satisfy $v_p(u-v) = 1$ and integral affine values M_u, M_v are congruent coefficientwise modulo p , then

$$\frac{B(u)M_u - B(v)M_v}{u-v} \in \mathbb{Z}_p[X] \quad (B \in \mathbb{Z}_p[t]).$$

This is the divided-difference step being used. When only one pole remains it is simpler. The condition $p^2 > 2K$ ensures that distinct ordinary nodes within a class differ with valuation exactly one.

For two indices $i, j < \ell - 2$, each residue has valuation at least

$$i + j + 6\delta - \ell - 4.$$

The row factors supply $i + j$, $W = D_N^5$ supplies 5δ , the node derivative loses $\ell - \delta - 1$, and a harmonic value loses at most 5. The polynomial contribution to A is integral. Hence valid *nonpositive* half-integer weights are

$$w_{a,i} = \begin{cases} \min(0, i + 3\delta - (\ell + 4)/2), & i < \ell - 2, \\ 0, & i \geq \ell - 2. \end{cases} \quad (4.12)$$

If $\delta = 1$, every early weight is obtained by truncating one of the half-integers $i - (\ell - 2)/2$. For two early rows the sum of the truncated weights is at most both zero and the displayed residue bound, as required. For an early and a late row, integrality of the whole entry suffices.

A single zero-class pole has weight $-1/2$. With two zero-class poles the entry bounds for the rows $1, t + p^2$ are $-3, -1, 0$, so $(-2, 0)$ are valid weights. We retain these source weights: replacing -2 by $-3/2$ would improve a bound, but such an optimization is not part of this edition.

Lemma 4.2 (Rank correction; source Lemma 4.2). *Let A, L be $h \times h$ matrices over a discretely valued field, or let A have polynomial entries equipped with the Gauss valuation. Suppose $v(A_{ij}) \geq w_i + w_j$, where every w_i is a nonpositive half-integer. If L is integral of rank at most r , and exactly z weights are zero, then*

$$v \det(A + p^{-1}L) \geq 2 \sum_i w_i - \min(r, z). \quad (4.13)$$

Stable target: Zeta5.rank_correction_bound.

Proof. Expand by complementary minors. A term using a $k \times k$ minor of L , on row set I and column set J , has valuation at least

$$2 \sum_i w_i - k - \sum_{i \in I} w_i - \sum_{j \in J} w_j.$$

If $k > r$ the minor vanishes. Order the weights $w_{(1)} \geq \dots \geq w_{(h)}$. The possible loss is at most

$$\max_{0 \leq k \leq \min(r, h)} \left(k + 2 \sum_{i=1}^k w_{(i)} \right).$$

This increases by one while $w_{(i)} = 0$ and has nonpositive increments thereafter, because a negative half-integer is at most $-1/2$. Its maximum is $\min(r, z)$. The argument uses only valuations and finite minors, so applies to the Gauss valuation as well. $\square$

4.3 Exact outer counts and large primes

Let

$$v = K - p \lfloor K/p \rfloor, \quad u = \max(0, N + v - p + 1), \quad t_p = \min(N, v) + u.$$

Define

$$\gamma_p^{\text{out}} = \begin{cases} -7(K - p) + 6t_p - 1 - \min(r_p, p - 1 - N + u), & K < 2p, \\ -7(K - p) + 3 + 12N + 5t_p - \min(r_p, p + u), & K \geq 2p. \end{cases} \quad (4.14)$$

Here are the finite counts behind this formula. For a removed class ($\delta = 1$), (4.12) gives

ℓ	2	3	4	5	6
$-2 \sum_i w_{a,i}$	0	1	2	4	6
$\#\{i : w_{a,i} = 0\}$	1	1	2	2	3

An unremoved class has cost $7(\ell - 2)$ and two zero weights. A zero class with one pole has cost 1 and no zero weight; with two poles it has cost 4 and one zero weight.

For $K < 2p$, an ordinary class has $\ell = 2 + e$, with $e = \mathbf{1}_{a \leq v} + \mathbf{1}_{a \geq p-v}$. In the removed classes $\sum e = t_p$, and the number with $e = 2$ is u . Their cost saves $6e$ relative to the unremoved formula. The total cost is $7(K - p) - 6t_p + 1$, and the total number of zero weights is $p - 1 - N + u$. For $K \geq 2p$, the same description is $\ell = 4 + e$. Each removed class saves $12 + 5e$, the number with $e = 2$ is again u , and the total cost and zero count are $7(K - p) - 3 - 12N - 5t_p$ and $p + u$. This proves the formula before the rank correction, and Lemma 4.2 adds its displayed minimum.

Proposition 4.3 (Outer valuation; source Proposition 4.3). *Under (4.9), $v_p^G(\Delta_K) \geq \gamma_p^{\text{out}}$. For the parameters (2.1), if $p > K$, then $v_p^G(\Delta_K) \geq 0$.*

Stable target: Zeta5.outer_determinant_valuation.

Proof. The first statement is (4.10), the class basis, (4.12), the exact counts, and Lemma 4.2. All rank and integrality properties of L survive the basis change.

For $p > K$, all integer pole labels are below p . Each ordinary square class has at most two poles, and there is no zero-class pole. Use Chinese remainders for these class factors, with local monic rows of successive degrees. The same congruence and divided-difference argument makes the pole contributions integral. The maximum degree of a polynomial quotient is $K + 4N - 2 < 2p - 3$, so all its moments are integral and the correction L vanishes. The transformed matrix is therefore integral, and its basis change is unimodular. $\square$

5 Normalization and the prime sum

For $p > K$ put $\gamma_p^{\text{out}} = 0$. For admissible K, M , define

$$L_p(K, M) = \begin{cases} -6h\ell_p(5K) - hv_p(24), & pM \leq K, \\ v_p(S_K) + \gamma_p^{\text{in}}, & pM > K, \ 3p \leq K, \\ v_p(S_K) + \gamma_p^{\text{out}}, & 3p > K. \end{cases} \quad (5.1)$$

The cases are disjoint and exhaustive because $M \geq 40$. Set

$$m_{K,M} = \prod_{p \leq 2h} p^{-L_p(K,M)} \in \mathbb{Q}_{>0}. \quad (5.2)$$

Negative and positive exponents are both retained. In particular this is not merely a denominator-clearing product with negative bounds truncated at zero. Legendre's formula supplies the exact exponents

$$\begin{aligned} v_p(S_K) &= 2h \sum_{a \geq 1} \lfloor K/p^a \rfloor - 12h \sum_{a \geq 1} \lfloor N/p^a \rfloor \\ &\quad - 2 \sum_{i=1}^{h-1} \sum_{a \geq 1} \lfloor 2i/p^a \rfloor + (h-1)v_p(4). \end{aligned} \quad (5.3)$$

All sums here are finite after their zero terms are removed.

Proposition 5.1 (Integrality; source Proposition 5.1). *For every admissible K, M , $Q_{K,M} = m_{K,M}F_K$ belongs to $\mathbb{Z}[X]$.*

Stable target: `Zeta5.normalized_polynomial_integral`.

Proof. For $p \leq 2h$, (3.13), Proposition 4.1, and Proposition 4.3 give $v_p^G(F_K) \geq L_p(K, M)$. Equation (5.2) therefore makes every coefficient integral at these primes. For $p > 2h$, all factorials in S_K are units and $p > K$, so Δ_K is integral by Proposition 4.3. A rational number integral at every prime is an integer: a prime divisor of its reduced denominator would otherwise give a negative valuation. Apply this observation to every coefficient. $\square$

5.1 The inner limiting functions

For $x \geq 3$ and $0 < z < 1/2$, put

$$\ell(x, z) = \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1, \quad b(x, z) = 3\ell(\alpha x, z).$$

Let

$$T = \lfloor 2Hx \rfloor, \quad s = Hx - T/2, \quad q = \lfloor 2x \rfloor, \quad n_+ = (2x - q)/2,$$

and define

$$\Gamma(x) = \int_0^{1/2} (T - b)(T + b - \ell - 5) dz + s(2T - q - 5) + (s - n_+)_+, \quad (5.4)$$

$$J(u) = mu - \frac{m(m+1)}{4}, \quad m = \lfloor 2u \rfloor, \quad (5.5)$$

$$\mathcal{N}(x) = 2\lambda x \lfloor x \rfloor - 12\lambda x \lfloor \alpha x \rfloor - 2J(\lambda x), \quad (5.6)$$

$$R(x) = -\Gamma(x) - \mathcal{N}(x). \quad (5.7)$$

In (5.4), $b = b(x, z)$ and $\ell = \ell(x, z)$. These functions are bounded and piecewise polynomial on every compact subinterval of $[3, \infty)$.

Lemma 5.2 (Uniform discrete-to-continuous inner estimate; source (5.7)). *For each fixed $M \geq 40$ there is a constant C_M such that, for every admissible K and every inner prime,*

$$|\gamma_p^{\text{in}} - p\Gamma(K/p)| \leq C_M, \quad |v_p(S_K) - p\mathcal{N}(K/p)| \leq C_M.$$

Stable target: `Zeta5.inner_uniform_limit`.

Proof. The finite counts satisfy $\ell_A(a) = \ell(A/p, a/p)$. On $3 \leq x \leq M$ the breakpoints in z of $\ell(x, z)$ and $\ell(\alpha x, z)$ form a uniformly bounded finite set. The number of grid points a/p in an interval differs from p times its length by at most two. All counts, dimensions, and weights involved are bounded by constants depending only on M .

The contribution of an ordinary class of dimension L and parameters b, ℓ to twice the sum of its weights is $L(L + 2b - \ell - 5)$. At the baseline $L = T - b$ this is the integrand of (5.4). Adding one row changes it by $2T - \ell - 4$. The counts ℓ are q or $q + 1$, and the set with the larger count has measure n_+ . Thus allocating the sp extra rows to the larger counts first gives, after division by p , $s(2T - q - 5) + (s - n_+)_+$.

Rounding the grid counts changes the available numbers in each category by $O_M(1)$. The reserved zero block, replacing $p/2$ by $(p - 1)/2$, and the $3m_N$ term in (4.4) change the ordinary row budget by $O_M(1)$. A change of one allocated row costs $O_M(1)$, so their total effect is $O_M(1)$; the zero block itself has that size as well. This argument remains uniform at an integer crossing of $2Hx$: giving one extra row to every ordinary class is exactly the next baseline allocation. It does not require a separation of $2Hx$ from the integers. There are finitely many count categories and bounded ranges of all these parameters for fixed M , so the indicated bounds give a single C_M .

For the scalar, $p^2 > 5K$ removes all higher prime powers from (5.3), and $p \geq 7$ removes the 4 term. Counting $\lfloor 2i/p \rfloor$ by its level sets gives

$$\sum_{i=1}^{h-1} \lfloor 2i/p \rfloor = pJ(h/p) + O_M(1).$$

For example, each nonempty level set has cardinality $h - jp/2 + O(1)$, and there are at most $2h/p = O(M)$ levels. Substitution in (5.3) proves the second assertion. $\square$

5.2 The outer limiting functions, with explicit indicator scope

Use $y = p/K$. Define

$$R_0(y) = \begin{cases} 8 - 9y - 8\alpha - 5 \min(\alpha, 1 - 2y) - 5(1 + \alpha - 3y)_+, & 1/3 < y < 1/2, \\ 7(1 - y) - 6 \min(\alpha, 1 - y) - 6(1 + \alpha - 2y)_+ + (1 + 4\alpha - 2y)_+, & 1/2 < y < 1, \\ 0, & y > 1. \end{cases} \quad (5.8)$$

Endpoint values may be assigned arbitrarily for integration. The rank correction is

$$d(y) = \begin{cases} (1 + 4\alpha - 3y - (1 + \alpha - 3y)_+)_+, & 1/3 < y < 1/2, \\ 0, & \text{otherwise.} \end{cases} \quad (5.9)$$

This is exactly source (5.9), with its *multiplicative* indicator written piecewise. In particular $\int_{1/3}^{1/2} d(y) dy = 9/640$.

Resolving (4.14) and using the fact that all integer endpoint corrections are bounded gives, uniformly for outer primes $p \leq K$,

$$-\gamma_p^{\text{out}} = K(R_0(p/K) - d(p/K)) + O(1), \quad (5.10)$$

$$-v_p(S_K) = K \left(-2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+ \right) + O(1). \quad (5.11)$$

The scalar formula holds throughout $1/3 < y \leq 2\lambda$; $N < p$ and $p^2 > 5K$ in this range. To verify uniformity directly, use $u = K(\alpha + 1 - (\lfloor 1/y \rfloor + 1)y)_+ + O(1)$ and $t_p = K \min(\alpha, 1 - \lfloor 1/y \rfloor y) + u$, then the Lipschitz property of minima and positive parts. The rank minimum in (4.14) is exactly what produces the subtraction of d .

Let

$$I_{\text{out}} = \int_{1/3}^{2\lambda} \left(R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+ \right) dy = \frac{127751}{96000}. \quad (5.12)$$

The exact piecewise evaluation is in Appendix B. For $1 < y < 2\lambda$, both residue terms vanish but the factorial scalar still contributes; this interval must not be omitted.

Lemma 5.3 (Weighted prime-sum consequence of the PNT). *Let $0 < a < b < \infty$, and let f be bounded and piecewise continuous on $[a, b]$. Then*

$$\frac{1}{K^2} \sum_{aK < p \leq bK} K f(p/K) \log p \longrightarrow \int_a^b f(y) dy.$$

Consequently, for bounded piecewise continuous g on $[3, M]$,

$$\frac{1}{K^2} \sum_{K/M < p \leq K/3} p g(K/p) \log p \longrightarrow \int_3^M g(x) x^{-3} dx.$$

Stable target: Zeta5.weighted_prime_sum.

Proof. Equation (1.1) implies $(\vartheta(vK) - \vartheta(uK))/K \rightarrow v - u$ for fixed $0 < u < v$. This proves the first statement for interval step functions. Upper and lower step approximations on a finite partition prove it for bounded Riemann-integrable functions, including bounded piecewise continuous ones. Bounds on the total measure follow from $\vartheta(bK)/K = O(1)$. A value at any one endpoint contributes at most $O(\log K/K)$, so endpoint conventions do not matter. For the second statement use $f(y) = yg(1/y)$ on $[1/M, 1/3]$ and the substitution $x = 1/y$. $\square$

Proposition 5.4 (Normalization growth; source Proposition 5.2). *For every fixed $M \geq 40$,*

$$\limsup_{\substack{K \rightarrow \infty \\ 40|K}} K^{-2} \log m_{K,M} \leq I_{\text{out}} + \frac{6\lambda}{M} + \int_3^M R(x) x^{-3} dx. \quad (5.13)$$

Stable target: Zeta5.normalization_growth.

Proof. Taking logarithms of (5.2) gives $\log m_{K,M} = -\sum_{p \leq 2h} L_p(K, M) \log p$, with signs unchanged. For $p \leq \sqrt{5K}$ the bound (3.13) contributes at most $O(K^{3/2} \log K) = o(K^2)$; the factors $v_p(24)$ involve only 2, 3. For $\sqrt{5K} < p \leq K/M$ the integer logarithm is one, and the PNT gives $6\lambda K^2/M + o(K^2)$.

In the inner range, Lemma 5.2 gives $-L_p = pR(K/p) + O_M(1)$. Its total error is bounded by $O_M(\vartheta(K)) = O_M(K) = o(K^2)$. Apply Lemma 5.3. The outer estimates (5.10)–(5.11) give (5.12) by the same lemma. All constants depending on M are used with M fixed before $K \rightarrow \infty$. $\square$

5.3 The periodic tail and the two fixed cutoffs

For $x \geq 3$ write

$$f = \{x\}, \quad g = \{\alpha x\}, \quad \tau = \{2Hx\}, \quad \sigma = \{2x\}, \quad \eta = \{2\lambda x\},$$

and let $A(z) = \ell(x, z) - 2x$, $B(z) = \ell(\alpha x, z) - 2\alpha x$. Expansion of (5.4)–(5.7) gives

$$R(x) = xF(x) + Q(x), \quad F(x) = 4\lambda + 2\lambda f - 12\lambda g. \quad (5.14)$$

$$Q(x) = \frac{\tau(\tau - \sigma) - (\tau - \sigma)_+ + \eta(1 - \eta)}{2} + 9 \int_0^{1/2} B(z)^2 dz - 3 \int_0^{1/2} A(z)B(z) dz. \quad (5.15)$$

Both A and B have integral zero on $(0, 1/2)$, and each squared integral is at most $1/8$. The mixed integral has absolute value at most $1/8$ by Cauchy–Schwarz. The numerator before the division by two in (5.15) is between $-1/4$ and $1/4$: if $\tau \leq \sigma$, its first two terms equal $\tau(\tau - \sigma) \in [-1/4, 0]$; otherwise they are $-(\tau - \sigma)(1 - \tau) \in [-1/4, 0]$, and $\eta(1 - \eta) \in [0, 1/4]$. Hence

$$-\frac{1}{2} \leq Q(x) \leq \frac{13}{8}. \quad (5.16)$$

An explicit exact verification of the expansion on every finite branch used below is supplied with Appendix B.

Define continuous periodic functions

$$\begin{aligned} \mathcal{P}(x) &= 74g(1 - g) - \lambda f(1 - f), & \overline{\mathcal{P}} &= \frac{2923}{240}, \\ G_0(v) &= v(1 - v)(2v - 1)/6, & \mathcal{C}(x) &= (74/\alpha)G_0(g) - \lambda G_0(f). \end{aligned}$$

Off their locally finite breakpoints, $\mathcal{P}' = F + \lambda$ and $\mathcal{C}' = \mathcal{P} - \overline{\mathcal{P}}$. They are continuous at those breakpoints, so integration by parts over subintervals telescopes without jump terms. Since $\max_{[0,1]} |G_0| = 1/(36\sqrt{3})$,

$$|\mathcal{C}(x)| \leq \frac{118511}{4320\sqrt{3}} < 16.$$

Two integrations by parts, first on a finite interval and then at infinity, give

$$\begin{aligned} \int_T^\infty \frac{R(x)}{x^3} dx &= -\frac{\lambda}{T} - \frac{\mathcal{P}(T)}{T^2} + \frac{\overline{\mathcal{P}}}{T^2} - \frac{2\mathcal{C}(T)}{T^3} \\ &\quad + 6 \int_T^\infty \frac{\mathcal{C}(x)}{x^4} dx + \int_T^\infty \frac{Q(x)}{x^3} dx. \end{aligned} \quad (5.17)$$

The boundary limits and absolute convergence follow from boundedness of $F, Q, \mathcal{P}, \mathcal{C}$.

At $T = 20$, $\mathcal{P}(T) = 37/2$ and $\mathcal{C}(T) = 0$, so

$$\int_{20}^{\infty} R(x)x^{-3} dx \leq -\frac{2689}{48000}. \quad (5.18)$$

If $40 \mid M$, then $\mathcal{P}(M) = \mathcal{C}(M) = 0$, and

$$\int_M^{\infty} R(x)x^{-3} dx \geq -\frac{\lambda}{M} + \frac{2923/240 - 1/4}{M^2} - \frac{32}{M^3}. \quad (5.19)$$

The finite integral is

$$\int_3^{20} R(x)x^{-3} dx = \frac{322437603634266857629}{7535670527041937280000}. \quad (5.20)$$

Put

$$A_* = \frac{9928298118277006344769}{7535670527041937280000}, \quad (5.21)$$

$$A_M = A_* + \frac{7\lambda}{M} - \frac{2923/240 - 1/4}{M^2} + \frac{32}{M^3}. \quad (5.22)$$

Combining Proposition 5.4 with (5.18)–(5.20) proves

$$\limsup_{\substack{K \rightarrow \infty \\ 40 \mid K}} K^{-2} \log m_{K,M} \leq A_M \quad (M \in 40\mathbb{Z}_{>0}). \quad (5.23)$$

Stable target: `Zeta5.normalization_quadratic_bound`.

6 The real determinant estimate

Define the external field on $[0, \infty)$ by

$$V(t) = 2\pi\sqrt{t} + \int_0^1 \log(t + u^2) du - 6 \int_0^\alpha \log(t + u^2) du. \quad (6.1)$$

The integrals at $t = 0$ are interpreted as their convergent improper integrals. For a finite measure ρ , write

$$U^\rho(t) = \int \log|t - u| d\rho(u), \quad I(\rho) = \iint \log|t - u| d\rho(t) d\rho(u),$$

whenever the integrals are absolutely convergent. The same notation is used on $\mathbb{C}$.

Proposition 6.1 (Explicit comparison measure; source Lemma 6.1). *There is a positive measure ρ of mass λ , supported in $(0, 2)$ and equal to the sum of the sixteen rationally specified arcsine measures in Appendix A, such that*

$$2U^\rho(t) - V(t) \leq M_0 := -\frac{1329}{200} \quad (t \geq 0). \quad (6.2)$$

Every supporting interval has length greater than $1/225$. With

$$C_* = -2\lambda + 12\alpha\lambda(1 - \log \alpha) + 3\lambda^2 - 2\lambda^2 \log(2\lambda), \quad (6.3)$$

$$U = -\frac{2733991}{2000000}, \quad (6.4)$$

one has

$$\lambda M_0 - I(\rho) + C_* < U. \quad (6.5)$$

Stable target: `Zeta5.comparison_measure_certificate`.

Proof. See Appendix A. The proof uses a finite rational partition with outward bounds for elementary functions. The independent checker supplied with this edition verifies all its cells and constants. This proposition uses only a comparison measure, not a solution or an existence theorem for the equilibrium-measure problem. $\square$

6.1 Zero-mass energy with explicit integrability

Lemma 6.2 (Zero-mass logarithmic energy; source Lemma 6.2). *Let ν be a finite real signed Borel measure on $\mathbb{C}$ with compact support, $\nu(\mathbb{C}) = 0$, and*

$$\iint |\log |z - w|| d|\nu|(z) d|\nu|(w) < \infty.$$

Here the absolute logarithmic kernel is assigned the value $+\infty$ on the diagonal $z = w$. In the signed real integral, use any representative on the resulting null diagonal. Then $I(\nu) \leq 0$.

Stable target: Zeta5. Analysis. zero_mass_logarithmic_energy.

Proof. For $s > 0$, completing the square gives

$$e^{-s|z-w|^2} = \frac{4s}{\pi} \int_{\mathbb{C}} e^{-2s|z-u|^2} e^{-2s|w-u|^2} dA(u),$$

where A is planar Lebesgue measure. Thus

$$J(s) := \iint e^{-s|z-w|^2} d\nu(z) d\nu(w) = \frac{4s}{\pi} \int_{\mathbb{C}} \left(\int e^{-2s|z-u|^2} d\nu(z) \right)^2 dA(u) \geq 0.$$

Fubini is justified by first integrating the nonnegative kernel against $|\nu| \otimes |\nu| \otimes A$; the completed-square identity bounds that integral by a constant times $|\nu|(\mathbb{C})^2$.

For $0 < a < b$ and $r > 0$, put

$$L_{a,b}(r) = \frac{1}{2} \int_a^b \frac{e^{-s} - e^{-sr^2}}{s} ds.$$

The identity $\frac{1}{2} \int_0^\infty (e^{-s} - e^{-sr^2}) s^{-1} ds = \log r$ follows by writing the numerator as $\int_1^{r^2} s e^{-sv} dv$ and integrating. The integrand has a fixed sign for fixed r , so $|L_{a,b}(r)| \leq |\log r|$ and $L_{a,b}(r) \rightarrow \log r$ as $a \downarrow 0, b \uparrow \infty$. Since ν has mass zero,

$$\iint L_{a,b}(|z - w|) d\nu(z) d\nu(w) = -\frac{1}{2} \int_a^b J(s) s^{-1} ds \leq 0.$$

The absolute logarithmic-integrability hypothesis implies that the diagonal has zero $|\nu| \otimes |\nu|$ measure. Dominated convergence therefore applies off that null set and proves the conclusion. $\square$

Lemma 6.3 (Circle regularization and arcsine error). *For $t_1, \dots, t_h \geq 0$ and $\varepsilon > 0$, let ω_i be normalized arclength on the circle of radius ε centered at t_i , and set $\sigma = K^{-1} \sum_{i=1}^h \omega_i$. Then*

$$\begin{aligned} 2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) &\leq (\lambda M_0 - I(\rho)) K^2 \\ &\quad + 120 \lambda K^2 \sqrt{\varepsilon} - h \log \varepsilon. \end{aligned} \tag{6.6}$$

Coincident points are allowed by assigning $-\infty$ to the left side.

Stable target: Zeta5. Analysis. regularized_configuration_bound.

Proof. The circle-average identity is

$$\int \log |z - u| d\omega_i(z) = \log \max(|t_i - u|, \varepsilon).$$

It follows by scaling the unit circle and integrating the power series for $\log(1 - z)$ inside its disk; the boundary case follows from the integrability of $\log |2 \sin(\theta/2)|$. This is also the standard circle-potential formula [3]. In particular $I(\omega_i) = \log \varepsilon$.

A unit arcsine measure on an interval of length L assigns at most $4\sqrt{2r}/(\pi\sqrt{L})$ to an interval of radius r . To see a sufficient bound, the maximum mass of an interval of fixed length occurs at an endpoint of the supporting interval; its mass is $2 \arcsin \sqrt{\min(2r/L, 1)}/\pi$, and $\arcsin u \leq 2u$ for $0 \leq u \leq 1$. Layer-cake integration then gives

$$\int U^\rho d\omega_i - U^\rho(t_i) \leq \sum_{j=1}^{16} \frac{8c_j \sqrt{2\varepsilon}}{\pi \sqrt{b_j - a_j}} \leq 60\sqrt{\varepsilon}.$$

The last step uses $b_j - a_j > 1/225$, $\sum c_j = \lambda < 1$, $\pi > 3$, and $\sqrt{2} < 3/2$.

The hypotheses of Lemma 6.2 need checking. The positive measure $\eta = \sigma + \rho$ has compact support. The positive part of the logarithmic kernel is bounded on that support. Circle potentials are bounded below by $\log \varepsilon$. For an arcsine measure on $[a, b]$, its potential at a complex point z is at least its potential at $\operatorname{Re} z$, because $|z - u| \geq |\operatorname{Re} z - u|$. The real arcsine potential is bounded below by $\log((b - a)/4)$. Consequently the negative logarithmic integral of η is uniformly bounded at points of its support, and Tonelli gives $\iint |\log |z - w|| d\eta(z) d\eta(w) < \infty$. Since $|\sigma - \rho| \leq \eta$ and both measures have mass λ , Lemma 6.2 applies to $\sigma - \rho$.

Applying the circle-average identity successively yields

$$\iint \log |z - w| d\omega_i(z) d\omega_j(w) \geq \log \max(|t_i - t_j|, \varepsilon) \geq \log |t_i - t_j|.$$

Therefore $K^2 I(\sigma) \geq 2 \sum_{i < j} \log |t_i - t_j| + h \log \varepsilon$. Expanding $I(\sigma - \rho) \leq 0$, bounding the mixed terms by the arcsine error, and applying (6.2) gives exactly (6.6). $\square$

A uniform configuration bound alone cannot be integrated on an unbounded domain. The next step explicitly leaves an integrable tail.

Lemma 6.4 (Retaining confinement). *For $K \geq 2$ and $t \geq 0$,*

$$2U^\rho(t) - V(t) + \sqrt{t}/K \leq M_0 + \sqrt{2}/K.$$

Consequently

$$\begin{aligned} 2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) + \sum_i \sqrt{t_i} &\leq (\lambda M_0 - I(\rho)) K^2 \\ &\quad + (120 + \sqrt{2})h + 2h \log K. \end{aligned} \tag{6.7}$$

Stable target: Zeta5. Analysis. confined_configuration_bound.

Proof. For $t \leq 2$ use (6.2). For $t \geq 2$, positivity and support of ρ imply $U^\rho(t) \leq \lambda \log t$. Also $\log(t + u^2) \leq \log t + u^2/t$. Thus

$$2U^\rho(t) - V(t) \leq \frac{13}{10} \log t + \frac{2\alpha^3}{t} - 2\pi\sqrt{t}. \tag{6.8}$$

After adding $\sqrt{t}/K$, this right side is decreasing for $t \geq 2$ and is less than M_0 at 2. Direct differentiation and the elementary bounds $\log 2 < 7/10$, $\pi > 3$, $\sqrt{2} > 7/5$ suffice. Apply the proof of Lemma 6.3 to the field $V(t) - \sqrt{t}/K$, using this bound and $\varepsilon = K^{-2}$. Since $h = \lambda K$, the regularization error becomes $120h$ and the self-energy term becomes $2h \log K$, which gives (6.7). $\square$

6.2 The determinant integral and all scaling factors

The finite determinant expansion and Fubini give

$$\Delta_K(\xi) = \frac{1}{h!} \int_{(0,\infty)^h} \prod_{i < j} (y_i^2 - y_j^2)^2 \prod_{i=1}^h \frac{D_N(y_i^2)^6}{D_K(y_i^2)} w(y_i) dy_i. \quad (6.9)$$

To prove this identity without importing a named determinant integration theorem, expand both Vandermonde determinants. The $h!$ choices of one permutation give identical moment-determinant expansions after relabeling integration variables. Every separated monomial integral is absolutely convergent by the exponential decay of w , which justifies Fubini and the finite rearrangements. Thus (6.9) is the required instance of Andréief's identity, with all its integrability hypotheses.

Stable target: `Zeta5.Analysis.gram_determinant_integral`.

The explicit weight bound

$$w(y) \leq 8192(1+y)^5 e^{-2\pi y} \quad (y > 0) \quad (6.10)$$

follows from the rational generating function for $\sum \ell^4 q^\ell$ and $1 - e^{-u} \geq u/(1+u)$. Indeed these give $w(y) \leq \pi^{-1}(1+2\pi y)^5 e^{-2\pi y}$, and $\pi < 4$ supplies $32\pi^4 < 8192$.

For $1 \leq m \leq K$ and $t \geq 0$, monotonicity gives

$$0 \leq \sum_{j=1}^m \log(t + (j/K)^2) - K \int_0^{m/K} \log(t + u^2) du \leq 2 \log K + 2. \quad (6.11)$$

For $t > 0$, differentiating the difference in t gives a nonpositive quantity because $u \mapsto (t + u^2)^{-1}$ is decreasing. At $t = 0$, the upper bound follows from $\log(m!) \leq m \log m - m + \log m + 1$. The continuous limit at zero justifies including that endpoint.

In (6.9) set $y_i = K\sqrt{t_i}$. The separate powers of K , before the Riemann-sum error, are

$$\begin{array}{ll} K^{2h(h-1)} & \text{from the Vandermonde,} \\ K^{(12N-2K)h} & \text{from } D_N^6/D_K, \\ K^{6h} & \text{from the weight and Jacobian.} \end{array}$$

The numerator error in (6.11) adds at most $K^{12h} e^{12h}$; the denominator uses the lower Riemann-sum bound. Consequently

$$\begin{aligned} \Delta_K(\xi) &\leq \frac{K^{2h(h+6N-K)+16h} (4096e^{12})^h}{h!} \\ &\quad \times \int_{(0,\infty)^h} \prod_{i < j} (t_i - t_j)^2 \prod_i e^{-KV(t_i)} t_i^{-1/2} (1 + \sqrt{t_i})^5 dt_i. \end{aligned} \quad (6.12)$$

There is no appeal to a uniform-bound-times-infinite-volume argument: use (6.7) and retain $\prod_i e^{-\sqrt{t_i}}$ in the integrand. Since

$$\int_0^\infty t^{-1/2} (1 + \sqrt{t})^5 e^{-\sqrt{t}} dt = 2 \sum_{j=0}^5 \binom{5}{j} j! = 652,$$

we obtain

$$\log \Delta_K(\xi) \leq 2h(h + 6N - K) \log K + (\lambda M_0 - I(\rho))K^2 + 18h \log K + 160h. \quad (6.13)$$

The linear constant bounds $\log 4096 + 12 + \log 652 + 120 + \sqrt{2}$ from above, and $h! \geq 1$.

Elementary factorial bounds and summation give

$$\begin{aligned} m \log m - m + 1 &\leq \log(m!) \leq m \log m - m + \log m + 2, \\ -2 \sum_{i=1}^{h-1} \log((2i)!) &\leq -2h^2 \log(2h) + 3h^2 + 4h \log(2h). \end{aligned}$$

Using (2.5) yields

$$\log S_K \leq (2\lambda - 12\alpha\lambda - 2\lambda^2)K^2 \log K + C_* K^2 + 6h \log K + 6h. \quad (6.14)$$

The leading coefficient of $K^2 \log K$ in (6.13) is $2\lambda(\lambda + 6\alpha - 1)$, exactly the negative of the coefficient in (6.14).

Proposition 6.5 (Real bound; source Proposition 6.3). *For every $K \in 40\mathbb{Z}_{>0}$,*

$$F_K(\xi) > 0, \quad \log F_K(\xi) \leq UK^2 + 24K \log K + 200K. \quad (6.15)$$

Stable target: `Zeta5.real_determinant_bound`.

Proof. Positivity is Proposition 2.3 and $S_K > 0$. Add (6.13) and (6.14), cancel the leading logarithmic terms, and use (6.5) and $h \leq K$. Enlarging the remaining linear coefficient to $200K$ gives the stated bound. $\square$

7 Completion of the polynomial estimates

Proof of Theorem 1.1. Proposition 5.1 proves integrality for every admissible pair. Proposition 2.2 proves degree h . Proposition 2.3 and positivity of both rational normalizations prove positive evaluation.

For fixed $M \in 40\mathbb{Z}_{>0}$, (5.23) and Proposition 6.5 imply

$$\limsup_{\substack{K \rightarrow \infty \\ 40|K}} K^{-2} \log Q_{K,M}(\xi) \leq A_M + U. \quad (7.1)$$

Appendix B proves, with exact positive margins,

$$-1600(A_{200} + U) > \frac{139}{5}, \quad -1600(A_{100000} + U) > \frac{7907}{100}.$$

Choose a positive slack strictly smaller than each margin after division by 1600. The definition of the limsup bound then gives a threshold for which the corresponding strict inequality holds for every allowed K . Increase that threshold to satisfy $K \geq 200M^2$, and put $K = 40n$. Since $K^2 = 1600n^2$, these are exactly (1.2)–(1.3). This establishes the theorem with existential, not numerically asserted, decay thresholds. $\square$

Theorem 1.3 now follows from the integer-polynomial criterion. No height estimate and no information about rational roots near ξ are needed for this main endpoint.

A The comparison measure and a reproducible finite certificate

A.1 Rational data and the arcsine formula

Let $\omega_{[a,b]}$ be the probability measure on (a, b) with density $1/(\pi\sqrt{(t-a)(b-t)})$, and put

$$\rho = \sum_{j=1}^{16} c_j \omega_{[a_j, b_j]}.$$

Table 1 specifies the data by integers with common denominator 10^{12} . Exact arithmetic gives

$$\sum_j c_j = 37/40, \quad 0 < a_{16} < \cdots < a_1 < b_1 < \cdots < b_{16} < 2, \quad b_j - a_j > 1/225.$$

The unit arcsine potential, with the present sign convention, is

Table 1: The rational arcsine measure (source Table 1).

j	$10^{12}a_j$	$10^{12}b_j$	$10^{12}c_j$
1	3906748086	8992695531	10515596180
2	2312248264	15340997855	29471737793
3	1402286665	25730180724	42934365099
4	881725356	41909578246	58204231966
5	578197906	65851089563	69037621310
6	396324613	99481037884	78873099189
7	283911191	144325727458	84856120711
8	212206188	201105762729	88396082127
9	165097686	269345996903	88303382125
10	133347132	347089554156	85472321255
11	111522114	430806704415	78899184238
12	96349355	515561896511	70353471918
13	85815639	595448778546	58838976615
14	78667711	664241383483	44421321106
15	74129565	716160577112	30462865791
16	71741310	746637295669	5959622577

$$U^{\omega_{[a,b]}}(t) = \begin{cases} \log((b-a)/4), & a \leq t \leq b, \\ \log\left(\frac{|t - (a+b)/2| + \sqrt{(t-a)(t-b)}}{2}\right), & t \notin [a, b]. \end{cases} \quad (\text{A.1})$$

In particular the square root and absolute value are inside the logarithm as displayed. To derive it, put $m = (a+b)/2$, $r = (b-a)/2$, and substitute $u = m + r \cos \theta$. For $t > b$, differentiation of the integral gives $((t-a)(t-b))^{-1/2}$; integrate and use $U(t) - \log t \rightarrow 0$ at infinity. Reflection treats $t < a$. For $t = m + r \cos \phi$ in the interval use $\cos \phi - \cos \theta = -2 \sin((\phi+\theta)/2) \sin((\phi-\theta)/2)$. The integral of the logarithms gives $\log(r/2)$, and continuity includes the endpoints.

With $S_j = \sum_{i \leq j} c_i$ and $S_0 = 0$, nesting gives

$$I(\rho) = \sum_{j=1}^{16} (S_j^2 - S_{j-1}^2) \log \frac{b_j - a_j}{4}. \quad (\text{A.2})$$

Indeed, the potential of component j is constant on every earlier support. Group its self-energy and both orders of its mixed energies; the coefficient is $c_j^2 + 2c_j S_{j-1} = S_j^2 - S_{j-1}^2$.

A.2 Elementary interval arithmetic and its soundness contract

For $1 \leq x \leq 2$, $z = (x - 1)/(x + 1)$, and $m \geq 1$,

$$0 \leq \log x - 2 \sum_{k=0}^{m-1} \frac{z^{2k+1}}{2k+1} \leq \frac{2z^{2m+1}}{(2m+1)(1-z^2)}. \quad (\text{A.3})$$

For general positive x , write $x = 2^e r$ with $1 \leq r < 2$ and use $\log x = e \log 2 + \log r$. For $0 \leq z \leq 1/2$,

$$\left| \arctan z - \sum_{k=0}^{m-1} (-1)^k \frac{z^{2k+1}}{2k+1} \right| \leq \frac{z^{2m+1}}{2m+1}. \quad (\text{A.4})$$

The reductions are

$$\arctan x = \frac{\pi}{2} - \arctan(1/x) \quad (x > 1), \quad \arctan x = 2 \arctan \frac{x}{1 + \sqrt{1+x^2}} \quad (0 \leq x \leq 1),$$

and $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. Equation (A.3) follows by integrating the geometric series for $2/(1-z^2)$ and bounding its positive tail. Equation (A.4) is the alternating-series estimate; the reductions follow from the tangent addition formula with their stated ranges fixing the branch. The Machin identity follows by the same addition formula and the bounds placing its value in $(3, 4)$.

The source uses $m = 64$ for logarithms, $m = 80$ for arctangents, and square-root brackets of width 2^{-144} . The independent checker retains the same series lengths and uses outward-rounded dyadic arithmetic at *192 bits*, including 192-bit square-root brackets. This changes the implementation precision, not any mathematical constant.

Here is the finite checker contract, intended to be proved once in Lean. An interval (l, u) represents $[l/2^{192}, u/2^{192}]$, with $l, u \in \mathbb{Z}$ and $l \leq u$. Addition and subtraction use endpoint arithmetic; multiplication encloses the four endpoint products, rounded outward. Reciprocal reverses the endpoints and is permitted only when zero is excluded. A nonnegative square root is enclosed by integers whose squares bracket the appropriately scaled rational endpoint. Each series is evaluated using these interval operations and enlarged by the explicit remainder bound above. Monotonicity allows logarithms and arctangents to be evaluated separately at the two rational endpoints. Induction over these operations proves enclosure soundness. All branch choices must be justified by exact endpoint comparisons.

The supplied `certificates/interval.py` implements these rules using integer arithmetic and rational numbers. Decimal output is used only for display. The complete accepted interval endpoints are stored in `certificates/verification_results.json`. Executing Python is not a kernel proof; a Lean checker must prove this soundness contract and replay the finite data or an equivalent exact computation.

Stable target: `Zeta5.Certificate.interval_soundness`.

A.3 The half-line potential inequality

For $t > 0$, direct integration in (6.1) gives

$$\begin{aligned} V(t) = & \log(1+t) - 6\alpha \log(t + \alpha^2) - 2 + 12\alpha \\ & + 2\sqrt{t} \left(\pi + \arctan \frac{1}{\sqrt{t}} - 6 \arctan \frac{\alpha}{\sqrt{t}} \right). \end{aligned} \quad (\text{A.5})$$

At zero, $V(0) = -12\alpha \log \alpha - 2 + 12\alpha$. If $\Phi(y) = V(y^2)$, then

$$\Phi'(y) = 2(\pi + \arctan(1/y) - 6 \arctan(\alpha/y)), \quad \Phi''(y) = -\frac{2}{1+y^2} + \frac{12\alpha}{\alpha^2 + y^2}.$$

The second derivative changes sign once, at $y^2 = 711/880$. Moreover $\Phi'(0+) = -3\pi$ and $\Phi'(+\infty) = 2\pi$. It follows that V has exactly one minimum: Φ' first increases from a negative value and then decreases to a strictly positive limit. Let

$$q_- = \frac{59205077}{10^{10}}, \quad q_+ = \frac{59205079}{10^{10}}.$$

The interval evaluation gives $\Phi'(\sqrt{q_-}) < 0 < \Phi'(\sqrt{q_+})$, so the minimizing t belongs to $[q_-, q_+]$. On that interval a lower bound is

$$\begin{aligned} V_* &= \log(1 + q_-) - 6\alpha \log(q_+ + \alpha^2) - 2 + 12\alpha \\ &\quad + 2\sqrt{q_+} \left(\pi + \arctan \frac{1}{\sqrt{q_+}} - 6 \arctan \frac{\alpha}{\sqrt{q_+}} \right). \end{aligned} \quad (\text{A.6})$$

The parenthesis is negative, as the same interval computation verifies. Bounding each logarithm and arctangent in its monotone direction and using this negative lower bound justifies the factor $\sqrt{q_+}$. In particular V_* is a lower bound for the global minimum of V .

The nested-support potential U^ρ is nonincreasing up to a_1 , constant on $[a_1, b_1]$, and nondecreasing thereafter. Thus its maximum on any interval is at an endpoint. For $[l, r] \subseteq [0, 2]$ define

$$B(l, r) = 2 \max(U^\rho(l), U^\rho(r)) - \begin{cases} V(r), & r \leq q_-, \\ V(l), & l \geq q_+, \\ V_*, & \text{otherwise.} \end{cases} \quad (\text{A.7})$$

The preceding monotonicity proves $2U^\rho(t) - V(t) \leq B(l, r)$ for $l \leq t \leq r$.

Define the ordered rational endpoints

$$A_0 = 0, \quad A_j = a_{17-j} \ (1 \leq j \leq 16), \quad A_{17} = q_-, \quad A_{18} = q_+, \quad A_{18+j} = b_j \ (1 \leq j \leq 16), \quad A_{35} = 2.$$

A row (j, d, k) in Table 2 denotes the closed cell

$$\left[A_j + (A_{j+1} - A_j) \frac{k}{2^d}, A_{j+1} + (A_{j+1} - A_j) \frac{k+1}{2^d} \right]. \quad (\text{A.8})$$

Every integer in a displayed inclusive range is used. The exact endpoint checks show that the 684 resulting cells cover $[0, 2]$ in order, with adjacent endpoints equal and with no overlap of interiors.

Table 2: The rational partition (source Table 2).

j	d	k (inclusive ranges)
0	1	0
0	2	2, 3
1	0	0
2	0	0

Continued on the next page

Table 2 continued

j	d	k (inclusive ranges)
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	1	0, 1
10	1	0, 1
11	1	0, 1
12	1	0
12	2	2, 3
13	2	0, ..., 3
14	2	0, ..., 3
15	1	0
15	2	2, 3
16	0	0
17	0	0
18	0	0
19	1	1
19	3	0, 2, 3
19	4	2, 3
20	2	3
20	3	4, 5
20	4	0, 1, 6, 7
20	5	4, ..., 11
21	2	3
21	3	5
21	4	0, 7, ..., 9
21	5	2, ..., 4, 10, ..., 13
21	6	10, ..., 19
22	3	5, ..., 7
22	4	0, 8, 9
22	5	2, 3, 12, ..., 15
22	6	8, ..., 23
23	3	6, 7
23	4	9, ..., 11
23	5	0, ..., 2, 14, ..., 17
23	6	6, ..., 13, 19, ..., 27
23	7	28, ..., 37
24	3	6, 7
24	4	10, 11
24	5	0, ..., 2, 15, ..., 19
24	6	6, ..., 11, 22, ..., 29
24	7	24, ..., 43
25	3	7
25	4	10, ..., 13
25	5	0, ..., 2, 16, ..., 19
25	6	6, ..., 11, 24, ..., 31
25	7	24, ..., 47
26	3	7

Continued on the next page

Table 2 continued

j	d	k (inclusive ranges)
26	4	11, ..., 13
26	5	0, ..., 2, 17, ..., 21
26	6	6, ..., 11, 25, ..., 33
26	7	24, ..., 49
27	4	11, ..., 15
27	5	0, ..., 2, 17, ..., 21
27	6	6, ..., 11, 26, ..., 33
27	7	24, ..., 51
28	4	12, ..., 15
28	5	0, ..., 2, 18, ..., 23
28	6	6, ..., 11, 27, ..., 35
28	7	24, ..., 53
29	4	13, ..., 15
29	5	0, ..., 2, 19, ..., 25
29	6	6, ..., 12, 27, ..., 37
29	7	26, ..., 53
30	4	13, ..., 15
30	5	0, ..., 2, 20, ..., 25
30	6	6, ..., 14, 26, ..., 39
30	7	30, ..., 51
31	4	14, 15
31	5	0, ..., 2, 20, ..., 27
31	6	6, ..., 39
32	4	15
32	5	0, ..., 4, 19, ..., 29
32	6	10, ..., 37
33	4	0, 15
33	5	2, ..., 29
34	2	2, 3
34	3	3
34	4	3, ..., 5
34	5	4, 5
34	6	5, ..., 7
34	7	6, ..., 9
34	8	7, ..., 11
34	9	5, ..., 13
34	10	0, ..., 9

For every listed cell the outward-rounded computation proves

$$B(l, r) < -\frac{6645002}{10^6} < -\frac{1329}{200}. \quad (\text{A.9})$$

For reproducibility, the largest certified upper endpoint occurred in cell $(j, d, k) = (30, 6, 26)$ and is smaller than $-664500268/10^8 = -6.64500268$. This is an upper bound on a cell bound, not a computation of the actual supremum of $2U^\rho - V$. For $t \geq 2$, (6.8) is decreasing and is already less than M_0 at 2. This proves (6.2) on the entire half-line.

Stable target: `Zeta5.Certificate.potential_cells`.

A.4 Energy, scalar constant, and the optional norm constant

The same interval rules applied to (A.2) and (6.3) give

$$-\frac{2126593445148}{10^{12}} < I(\rho) < -\frac{2126593445147}{10^{12}}, \quad (\text{A.10})$$

$$\frac{2653035990340}{10^{12}} < C_* < \frac{2653035990341}{10^{12}}. \quad (\text{A.11})$$

Since $\lambda M_0 = -49173/8000$, it follows that

$$\lambda M_0 - I(\rho) + C_* < -\frac{1366995564511}{10^{12}} < -\frac{2733991}{2000000} = U.$$

This completes Proposition 6.1. For the constant κ in (C.5) below, the checker also proves

$$\frac{2773171335773}{2000000000000} < \kappa < \frac{6932928339433}{5000000000000} < \frac{6933}{500}. \quad (\text{A.12})$$

Stable target: `Zeta5.Certificate.energy_and_norm_constants`.

B Exact arithmetic integrals and margins

B.1 A branch-complete calculation of the inner integral

For $0 \leq u < 1$, define

$$d_0(u) = \min(u, 1 - u), \quad e(u) = \begin{cases} 1, & u \leq 1/2, \\ -1, & u > 1/2. \end{cases}$$

For real x and almost every $z \in (0, 1/2)$,

$$\ell(x, z) - 2x = e(\{x\})(\mathbf{1}_{z < d_0(\{x\})} - 2d_0(\{x\})).$$

The exceptional equality case in z is a singleton. Thus, with f, g, A, B as in Section 5,

$$\int_0^{1/2} B^2 = d_0(g)(1 - 2d_0(g)), \quad (\text{B.1})$$

$$\int_0^{1/2} AB = e(f)e(g)(\min(d_0(f), d_0(g)) - 2d_0(f)d_0(g)). \quad (\text{B.2})$$

These follow by integrating indicator functions and their product; for example the intersection has length $\min(d_0(f), d_0(g))$.

Take the ordered set consisting of 3, 20, and every $x \in (3, 20)$ for which $cx \in \mathbb{Z}$ for some

$$c \in \{2, 2\alpha, 2\lambda, 2H, 4\alpha, 2(1 - \alpha), 2(1 + \alpha)\}. \quad (\text{B.3})$$

There are exactly 143 open intervals (t_i, t_{i+1}) . On each, all floors, signs, and branches of minima and positive parts are fixed. The extra possible crossings $d_0(f) = d_0(g)$ and $\tau = \sigma$ are included in (B.3). The quadratic coefficient in R cancels because

$$2\lambda - 12\lambda\alpha + 2(H^2 - H - \lambda^2) - 18\alpha^2 + 6\alpha = 0.$$

Therefore $R(x) = a_i x + b_i$ on each such open interval, and

$$\int_3^{20} R(x) x^{-3} dx = \sum_{i=0}^{142} \left[a_i \left(\frac{1}{t_i} - \frac{1}{t_{i+1}} \right) + \frac{b_i}{2} \left(\frac{1}{t_i^2} - \frac{1}{t_{i+1}^2} \right) \right]. \quad (\text{B.4})$$

Values at finitely many endpoints do not affect the integrals.

The independent verifier does not merely sample two points and assume linearity. It forms exact rational polynomials on each branch, verifies every branch inequality at both endpoints, and computes Γ from its original integral on the strips in z bounded by $0, d_0(f), d_0(g), 1/2$. It separately forms (5.14)–(5.15) using (B.1)–(B.2), checks equality of all polynomial coefficients, and checks cancellation of degrees greater than one. Then it integrates the resulting affine polynomial exactly by (B.4). The 143 coefficient records are included in the JSON output. Grouping them by integer unit intervals gives Table 3; its sum is (5.20).

Stable target: `Zeta5.Certificate.inner_integral`.

Table 3: Inner integral on unit intervals (source Table 3).

j	$\int_j^{j+1} R(x) x^{-3} dx$
3	$\frac{26807}{161280}$
4	$\frac{37383}{704000}$
5	$\frac{37523}{8236800}$
6	$-\frac{120923}{6552000}$
7	$-\frac{10025233}{356428800}$
8	$-\frac{110029309}{3348864000}$
9	$-\frac{2278419487}{64465632000}$
10	$-\frac{282415081}{7724640000}$
11	$-\frac{3236921227}{87524236800}$
12	$-\frac{3350220001}{90899827200}$
13	$\frac{7424224373}{2217983040000}$
14	$\frac{16775764609}{955086612480}$
15	$\frac{369043847}{30810528000}$
16	$\frac{651380108633}{86461373856000}$
17	$\frac{472851276229}{119820121344000}$
18	$\frac{4930060867}{4724197793280}$
19	$-\frac{15199801}{11563552000}$

B.2 The outer integral

Let

$$T(y) = R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+.$$

On each open interval of Table 4, $T(y) = b + cy$. Resolving every minimum, positive part, and reciprocal floor by exact endpoint comparisons proves those affine expressions. Their integrals are $b(r - l) + c(r^2 - l^2)/2$, and their sum is $I_{\text{out}} = 127751/96000$. As before, no assertion about an affine branch is inferred from numerical sampling.

Stable target: `Zeta5.Certificate.outer_integral`.

Table 4: Outer integrand $T(y) = b + cy$ (source Table 4).

l	r	b	c
$\frac{1}{3}$	$\frac{43}{120}$	$\frac{279}{40}$	-9
$\frac{43}{120}$	$\frac{37}{100}$	$\frac{451}{40}$	-21
$\frac{37}{100}$	$\frac{13}{30}$	$\frac{377}{40}$	-16
$\frac{13}{30}$	$\frac{37}{80}$	$\frac{429}{40}$	-19
$\frac{37}{80}$	$\frac{1}{2}$	$\frac{17}{4}$	-5
$\frac{1}{2}$	$\frac{43}{80}$	$\frac{51}{10}$	-3
$\frac{43}{80}$	$\frac{37}{60}$	$\frac{231}{20}$	-15
$\frac{37}{60}$	$\frac{13}{20}$	$\frac{97}{10}$	-12
$\frac{13}{20}$	$\frac{37}{40}$	$\frac{42}{5}$	-10
$\frac{37}{40}$	1	1	-2
1	$\frac{37}{20}$	$\frac{37}{20}$	-1

B.3 The final rational margins

Direct substitution into (5.22) gives

$$A_{200} = \frac{127125602969131786927559}{94195881588024216000000},$$

$$A_{100000} = \frac{7756864096839411316755964319057}{5887242599251513500000000000000}.$$

With $U = -2733991/2000000$, the exact differences are

$$-1600(A_{200} + U) - \frac{139}{5} = \frac{3089837638249482469}{58872425992515135000} > 0, \quad (\text{B.5})$$

$$-1600(A_{100000} + U) - \frac{7907}{100} = \frac{29873543950273155160680943}{3679526624532195937500000000} > 0. \quad (\text{B.6})$$

These imply all strict inequalities used in Section 7. In particular $A_{100000} + U < -79/1600$.

Stable target: `Zeta5.Certificate.final_rational_margins`.

C Optional rational approximation and coefficient height

This appendix retains source Appendix C. Its matrix norm bound is not an assumption of Theorem 1.3.

C.1 Relative norm of the coefficient matrix

Put

$$G = G_K(\xi), \quad M_K = [X]G_K(X), \quad B_K = G^{-1/2}M_KG^{-1/2}.$$

The square root and inverse are the real symmetric positive-definite ones supplied by Proposition 2.3. Norm means Euclidean operator norm.

Proposition C.1 (Relative norm; source Proposition C.1). *For all sufficiently large $K \in 40\mathbb{Z}_{>0}$,*

$$\|B_K\| \leq e^{\varrho K}, \quad \varrho = \frac{6933}{500}. \quad (\text{C.1})$$

Stable target: `Zeta5.Approx.relative_matrix_norm`.

Proof. Let q be a real polynomial of degree less than h and put $p(t) = q(K^2t)$ and $A_K = K^{12N-2K+6}$. For any $0 < a < b$, keep only the $\ell = 1$ term of the positive weight, restrict $y = K\sqrt{t}$ to $a \leq t \leq b$, and bound every factor separately. This gives

$$\mathbf{q}^\top G \mathbf{q} \geq A_K c_a e^{-F(a,b)K} \int_a^b p(t)^2 dt, \quad (\text{C.2})$$

where

$$c_a = \frac{(2\pi)^4 a^2}{24}, \quad F(a,b) = 2\pi\sqrt{b} + \log(1+b) - 6\alpha \log a.$$

The scaled numerator is at least a^{6N} , the denominator is at most $(1+b)^K$, and the remaining scaled density is at least $c_a e^{-2\pi K\sqrt{b}}$. This proves the bound with its exact power of K .

Write $u_j = j/K$. The residue expression in Proposition 2.2 gives

$$\mathbf{q}^\top M_K \mathbf{q} = A_K \sum_{j=N+1}^K r_j p(-u_j^2)^2, \quad r_j = \frac{u_j^4 \prod_{\ell=1}^N (u_\ell^2 - u_j^2)^5}{\prod_{N < k \leq K, k \neq j} (u_k^2 - u_j^2)}. \quad (\text{C.3})$$

The numerator has absolute value at most one. Since $u_k + u_j \geq 2\alpha$,

$$\begin{aligned} |r_j| &\leq \frac{(K/(2\alpha))^{h-1}}{(j-N-1)!(K-j)!} \leq \frac{(K/\alpha)^{h-1}}{(h-1)!} \\ &\leq \exp\{\lambda K(1 - \log(\alpha\lambda))\}. \end{aligned}$$

The middle inequality is the bound $\binom{h-1}{j-N-1} \leq 2^{h-1}$. For the last one, use $m! \geq (m/e)^m$ and monotonicity of $x \log(eK/(\alpha x))$ for $0 < x \leq h$.

Map $[a, b]$ to $[-1, 1]$ and expand p in the orthogonal Legendre basis. For $-1 \leq t \leq 0$ the transformed argument has absolute value at most

$$Y = \frac{2+a+b}{b-a} > 1, \quad R = Y + \sqrt{Y^2 - 1}.$$

The Laplace integral for the Legendre polynomial gives $|P_j(s)| \leq R^j$ on this exterior interval (use parity for negative s). Cauchy–Schwarz and $\sum_{j=0}^{h-1} (2j+1) = h^2$ imply

$$\max_{-1 \leq t \leq 0} |p(t)|^2 \leq \frac{h^2 R^{2h}}{b-a} \int_a^b p(t)^2 dt. \quad (\text{C.4})$$

Combining (C.2)–(C.4) and using the Rayleigh quotient for the real symmetric matrix B_K gives

$$\|B_K\| \leq \frac{h^3}{c_a(b-a)} e^{\kappa(a,b)K}, \quad \kappa(a,b) = F(a,b) + \lambda(1 - \log(\alpha\lambda)) + 2\lambda \log R.$$

Choose $a = 1/18$, $b = 1/3$, so $Y = 43/5$ and

$$\begin{aligned} \kappa &= \frac{2\pi}{\sqrt{3}} + \log \frac{4}{3} + \frac{9}{20} \log 18 + \frac{37}{40} \left(1 + \log \frac{1600}{111} \right) \\ &\quad + \frac{37}{20} \log \frac{43 + 4\sqrt{114}}{5} < \frac{6933}{500}. \end{aligned} \quad (\text{C.5})$$

The strict margin is certified in (A.12). Since a fixed polynomial factor in h is eventually at most $e^{(\varrho - \kappa)K}$, it is absorbed by this positive margin. This proves (C.1). $\square$

C.2 The approximation contradiction with explicit quantifiers

Proof of Corollary 1.4. Fix $M = 100000$ and write $Q_K = Q_{K,M}$. From (7.1) and Appendix B, for all sufficiently large allowed K ,

$$0 < Q_K(\xi) \leq e^{-\epsilon K^2}, \quad \epsilon = 79/1600.$$

Let $c = 75/4$. For an integer $b \geq 2$, choose

$$K = 40 \left\lceil \frac{c \log b}{40} \right\rceil, \quad c \log b \leq K < c \log b + 40.$$

As $b \rightarrow \infty$, this K satisfies admissibility and all eventual bounds above. Suppose $a \in \mathbb{Z}$ and $\delta = a/b - \xi$ satisfies $|\delta| \leq b^{-260}$. The exact rational identities

$$\varrho c - 260 = -1/80, \quad \epsilon - \lambda/c = 1/24000$$

give $|\delta| \|B_K\| \leq e^{40\varrho} b^{-1/80} = o(K^{-1})$. All eigenvalues of $I + \delta B_K$ are therefore positive for sufficiently large b , uniformly in a , and

$$0 < \det(I + \delta B_K) \leq (1 + |\delta| \|B_K\|)^h \leq e^{h|\delta| \|B_K\|} \rightarrow 1.$$

It is eventually at most two. Affine dependence on X gives the exact identity $Q_K(a/b) = Q_K(\xi) \det(I + \delta B_K)$. Integrality and degree h now imply that $b^h Q_K(a/b)$ is a positive integer, whereas

$$b^h Q_K(a/b) \leq 2 \exp(\lambda K \log b - \epsilon K^2) \leq 2e^{-K^2/24000} < 1$$

for every sufficiently large b . This contradiction proves the corollary with a single threshold b_0 independent of a . $\square$

C.3 Coefficient height and primitive normalization

If $H(Q)$ denotes the largest absolute value of an integer coefficient, the real eigenvalues $\beta_1, \dots, \beta_h$ of B_K give

$$Q_K(X) = Q_K(\xi) \prod_{j=1}^h (1 + (X - \xi)\beta_j).$$

Since $\xi < 2$, the product of the coefficient ℓ^1 norms bounds the height, and Proposition C.1 gives

$$\log H(Q_K) \leq (\lambda\varrho - \epsilon)K^2 + o(K^2) = \frac{511067}{40000}K^2 + o(K^2) < 13K^2 \quad (\text{C.6})$$

for sufficiently large allowed K . For instance, each factor has coefficient sum at most $1 + 3e^{\varrho K}$, which proves the required $O(K)$ remainder after multiplication by h .

Stable target: `Zeta5.Approx.coefficient_height`.

For a nonzero $A \in \mathbb{Q}[X]$, let $\text{cont}_{\mathbb{Q}}(A) > 0$ be its rational content: $A/\text{cont}_{\mathbb{Q}}(A)$ is integral and primitive. Set

$$P_K = \Delta_K / \text{cont}_{\mathbb{Q}}(\Delta_K).$$

For every admissible K, M there is an integer $a_{K,M} \geq 1$ with $Q_{K,M} = a_{K,M}P_K$. To see this, a rational multiple of a primitive integer polynomial is integral only if the scalar is an integer, by Bézout applied to its coefficients. Here the scalar is positive. Thus $0 < P_K(\xi) \leq Q_{K,M}(\xi)$, and the established upper bounds also hold for the primitive polynomial. The empirical primitive values reported in the supplied audits are not needed for this deduction.

Stable target: `Zeta5.primitive_normalization`.

D Autonomous Lean formalization specification

D.1 The endpoint and the trust boundary

The accompanying declaration ledger is an implementation blueprint, not a list of already formalized results. At the time of this edition no Lean proofs in this package have been checked. The finite Python verification has been executed successfully, but it certifies only the displayed finite arithmetic and interval enclosures, not the asymptotic or local functional theorems.

Use a named real definition

$$\text{zetaFive} := \sum_{n=0}^{\infty} \frac{1}{(n+1)^5}$$

with real coercions and a summability proof. The final unconditional statement must be **Irrational zetaFive**, together with a theorem identifying it with the intended $\zeta(5)$ value. A theorem about an abstract real parameter satisfying the polynomial estimates is the useful conditional endpoint, not a substitute for this theorem.

During construction, a structure of named proof inputs is acceptable. Its fields must be explicit propositions, not project axioms, and the conditional theorem must retain that structure as a visible argument. Standard external theorems must likewise be identified by exact type and provenance. An unconditional endpoint may be exported only after every input has been instantiated by checked proofs or actual imported theorems with audited dependencies. No incomplete definition, **sorry**, **admit**, false premise, or added axiom may silently close an obligation.

The final certificate path must be kernel-checkable. An external Python program may emit rational witnesses and subdivision data; the theorem asserting that they imply the desired inequalities must be proved in Lean. Unchecked numerical or native execution is not by itself a proof of a certificate. Record any additional computational trust dependency rather than hiding it. Inspect `#print axioms` for every public endpoint and retain the output in the build audit; see the Lean reference manual for axiom tracking and validation [5].

D.2 Minimal mathematical interfaces for external inputs

The following interfaces state the mathematical content to be found in libraries or proved; they do not guess current declaration names.

Bernoulli arithmetic.

$B_1 = -1/2$; the polynomial difference, reflection, and multiplication identities; $B_{2k+1} = 0$ for $k \geq 1$; and, for primes p and $m \geq 1$, $v_p(B_{2m}) \geq -1$ with p -integrality if $p - 1 \nmid 2m$. The von Staudt–Clausen denominator statement is a sufficient interface.

Even zeta and Hermite.

Euler’s formula at positive even integers, the identification $\zeta(5, j) = \xi - H_{j-1}^{(5)}$ for integers $j \geq 1$, and, for $a > 0$,

$$\zeta(5, a) = \frac{1}{2a^5} + \frac{1}{4a^4} + 2 \int_0^\infty \frac{\operatorname{Im}(a - iy)^{-5}}{e^{2\pi y} - 1} dy,$$

including absolute integrability of the integral. An independently proved equivalent pole-moment formula is also acceptable if its constants and domain are shown to match Definition 2.1.

Prime number theorem.

Equation (1.1). Prove the weighted finite-interval consequence as Lemma 5.3; do not assume a stronger normalization bound without a bridge to the actual exponents L_p .

Integration and elementary functions.

Complete valued fields, uniform convergence of restricted power series, Bochner integration/Fubini/dominated convergence in the exact domains used above, elementary gamma integrals, change of variables on the positive half-line, and the logarithm/arctangent identities used in the finite checker. The zero-mass energy and determinant integral have proofs in Section 6 and remain explicit project targets.

Optional Legendre interface.

Orthogonality with squared norm $2/(2j + 1)$ on $[-1, 1]$; parity; and the exterior bound $|P_j(s)| \leq (|s| + \sqrt{s^2 - 1})^j$ for $|s| \geq 1$. These support only Appendix C.

D.3 Representation and statement-level hazards

1. Work with the integer parameter n and set $K = 40n$, $N = 3n$, $h = 37n$. This avoids accidental natural-number division or unproved divisibility. Represent $Q_{K,M}$ on an admissible parameter subtype, or supply harmless values outside the admissible domain and prove the bridge on that domain.

2. Define the rational functional using its unique polynomial and simple-pole decomposition. The domain is a vector space; arbitrary products may introduce double poles and leave the domain. Extend scalars explicitly before applying it over $\mathbb{Q}_p$.
3. Use a restricted sequence space plus a finite residue vector if that avoids a general Tate-algebra library. Preserve the bounded analytic remainder and the near/far distinction. The target of a convergent local evaluation has bounded polynomial degree; do not silently use completeness of all of $\mathbb{Q}_p[X]$ in the Gauss norm.
4. State all arithmetic bounds for the *Gauss valuation of coefficients*, including the constant coefficient, not merely for the specialized value at ξ . Keep the small primes 2, 3, 5 in the universal argument and retain the factor $v_p(24)$.
5. Keep the signs of L_p and use integer powers in (5.2). The inner row weights may be positive. Only the rank-correction lemma requires all its weights to be nonpositive. Its rank hypothesis is rank over $\mathbb{Q}_p$, not rank of a reduction modulo p .
6. Keep M fixed before passing to $K \rightarrow \infty$. Distinguish the explicit admissibility threshold from the eventual asymptotic threshold. A numerical check at $n = 200000$ is not a proof of eventual decay.
7. In standard Lean real analysis, `Real.log 0` is a totalized value, not $-\infty$ (see [6]). The logarithmic-energy interface must therefore include nullity of the diagonal as well as integrability of a real-valued representative, or use an extended-valued singular kernel and prove that nullity. For configuration bounds first assume the points are pairwise distinct; treat collisions separately by the zero Vandermonde product. Do not translate the displayed $-\infty$ convention into a false real inequality involving a totalized logarithm.
8. Before every inverse, logarithm, and determinant quotient, retain the needed nonzero or positivity proof. The source weight is defined for $y > 0$; a value at $y = 0$ must be assigned separately if a globally continuous extension is desired.
9. Validate every certificate branch, including square-root domains, reciprocal nonvanishing, partition coverage, and the finite set of excluded integral endpoints. Validating only final floating-point values is insufficient. In the outer certificate retain the entire interval $(1, 2\lambda)$ and the piecewise definition (5.9).

D.4 Recommended build stages and completion criteria

Stage	Module group	Deliverable
1	Definitions and endpoint	Actual determinant definitions, real-series target, rational-evaluation lemma, and a visibly conditional end-to-end theorem.
2	Exact arithmetic	Rational constants, parameter identities, Appendix B branch certificates, and a sound interval checker.
3	Completed local functional	Bounded division, principal-part compatibility, reflection/difference, and the full distribution identity.
4	Local determinant bounds	Small-prime bound, both unimodular bases, all weight comparisons, field-rank correction, integrality.
5	Prime normalization	Uniform finite-count estimates, PNT interface, weighted prime sums, periodic-tail bound, fixed-cutoff growth.
6	Positive and real analysis	Moment representation, Gaussian zero-mass energy, regularization, confinement, determinant integral.
7	Main endpoint	Instantiate every input; prove integer polynomial decay and the unconditional irrationality theorem; audit axioms.
8	Optional consequences	Relative norm, exponent 260, height, primitive normalization.

The stages may proceed in parallel where dependencies allow. The local functional and numerical certificates deserve early attention because they resolve the most compressed parts of the source argument.

The files `formalization/declarations.csv` and `formalization/dependency_graph.md` give the stable target list and its dependencies. `AGENTS.md` supplies operational instructions. No proposed theorem may be weakened silently to make a build pass. When an obstruction occurs, record the exact failed statement and a minimal counterexample or missing lemma; distinguish a mathematical counterexample from missing library infrastructure. Keep public status conditional until the endpoint has no project-owned assumptions.

D.5 Verification performed for this edition

The finite checker was executed with ordinary Python integer arithmetic and 192-bit outward-rounded intervals. It accepted all sixteen measure components, the minimizing-point bracket and its sign conditions, all 684 potential cells, the energy and scalar enclosures, the optional κ enclosure, the 143 symbolic inner branches, all seventeen unit integrals, the eleven outer branches, and both exact final margins. It also checks the elementary rational tail calculation at 20 and the two approximation-constant identities.

The provided audits additionally report small- K determinant calculations and truncated p -adic distribution tests. Those reports are preserved as provenance in `REPAIRS.md`; they have not been relabelled as computations performed in this edition, and none is used as a proof premise. This document does not assert that all possible mathematical defects have been excluded. It makes the revised proof obligations explicit and supplies reproducible finite certificates for the computations it actually checks.

References

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- [2] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Chapters 18, 24, 25, and 27. In particular (18.10.5), (24.4.1), (24.4.3), (24.4.18), §24.10(i), (25.6.2), (25.11.29), and §27.12. <https://dlmf.nist.gov/>.
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- [6] The Mathlib documentation, `Mathlib.Analysis.SpecialFunctions.Log.Basic`, in particular `Real.log_zero`, consulted 23 September 2026. https://leanprover-community.github.io/mathlib4_docs/Mathlib/Analysis/SpecialFunctions/Log/Basic.html.